

FINAL PROJECT

CM 3070

DRAFT REPORT

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Abstract

Automated Essay Scoring is a rapidly developing research areas in NLP with respect to education. According to [1] this is due to the rapid progress in Artificial Intelligence which has created new possibilities. Some of the benefits of AES are enhanced fairness, consistency and efficiency in the scoring process [2]. This research has shown Optical Character Recognition (OCR) technology has the potential to provide accessibility to an AES system by allowing essays to be submitted in handwritten format as images. In limited experimentation with only two handwritten samples, PaddleOCR achieved average CER and WER scores of 0.771 and 0.962 respectively. This research investigates whether orchestrating OCR and ASR models along with a Transformer can produce an AES system this interpretable, accessible and reliable.

Chapter 1

Introduction

The Proposed System

BrainPower- A hybrid, multi-modal, Automated Essay Scoring (AES) system which orchestrates an OCR, STT and Transformer model, providing accessibility for students and interpretability, readability and automation in support of human scorers.

The template selected

Template 4.1 Project Idea 1: Orchestrating AI models to achieve a goal

Rationale for topic selection

Reading and providing corrective feedback on a single 250-word essay can take 10-15 minutes on average and a typical English A class has 35 students. This manual approach does not scale well if a teacher is assigned to more than one class. As an educator, the author recognizes the stressful, time-consuming ordeal that educators endure when marking answers to open-ended questions, especially essay. Currently accessibility for students who can speak but are incapable of writing or typing due to disability is provided by means of a scribe. Although well intentioned, this arrangement demands extra human resources, is time consuming and can be intimidating to some students. It would be more beneficial to the student if they could submit an answer to a prompt independently by verbally expressing their ideas recording, transcribing and then having them evaluated. There should also be a mechanism in place to allow students who are unable to be present at school to be able to under supervision write essays and share photographs or scan copies of these essays.

Chapter 2

Literature Review

2.1.0 Automated Essay Scoring (AES) Systems

Key terms: traits, features, handmade features, engineered features, holistic score, neural networks, transformers, prompts, QWK - QWK (Quadratic weighted Kappa) evaluates the degree of agreement among human and AI scorers, it is a metric between 0 and 1, representing no agreement beyond chance and perfect agreement, respectively [1].

This section is designed to give the user a historical perspective on AES system describing their advantages and limitations and, in the process, justify the selection of PaddleOCR, Whisper and DeBERTa in the AES pipeline, to evaluate and score handwritten, typed and spoken essays.

What is AES?

According to [2], automated essay scoring refers to the process of grading student essays without human interference. An AES system takes as input an essay written for a given prompt (the topic) and then assigns a numeric score to the essay reflecting its quality, based on traits such as content, grammar and organization.

Early AES Systems-Handcrafted feature system

According to [3], early AES systems such as Project Essay Grader (PEG), IEA, E-rater, IntelliMetric and BETSY, all used handcrafted feature approaches. Handcrafted features are values that are encoded into feature vectors, for example, the first dimension of a vector may represent word count and the second, sentence length and so on. These are then fed to a neural network for processing. This approach gives features positional significance, resulting in an interpretable output.

According to [4], these features are meticulously designed by researchers based on linguistic expertise, allowing for adaptability to various scoring criteria. [4], although stating the fact that these systems were exceptionally accurate, noted that despite the reliance on handcrafted features, conventional AES systems demonstrate robust performance even with limited datasets.

Limitations:

[2] stated that the process of feature engineering is the most difficult part of building an

AES system, and that it is also challenging for humans to consider all the factors that are involved in assigning a score to an essay. There is also no evidence to suggest that any of these earlier systems graded handwritten essays.

Feature engineered systems

Neural approaches engineer features like sentence length and number of paragraphs and encode these in dense numerical feature vectors. A dense vector can contain several numerical values relating to a feature or combination of features. These numbers have no positional significance to humans as compared to handcrafted features as they are assigned based on weight adjustments influenced by gradient descent. This enables the network to iteratively adjust weights to learn a model that can then generalise to score new essays.

[2] implemented a model for holistic scoring based on recurrent neural networks (LSTM), to learn the relationship between an input essay directly and its assigned score. The system encodes the information required for essay evaluation and learns the complex patterns in the data. Their system outperformed a strong baseline by 5.6% in terms of QWK, without requiring any feature engineering.

Limitations: The system was not tested on a handwritten dataset, especially not one with cursive handwritten essays. According to [5], neural approaches provide scores without giving evidence of how they are arrived at. Since one of the goals of BrainPower is to produce interpretable results, a neural approach will not be employed.

Furthermore, more modern approaches such as transformers have been shown to be superior to neural approaches. [6] compared the performance of three neural approaches LSTM, Bi-LSTM and BERT a transformer model and concluded that the transformer model offered superior performance. [5] reviewed several pretrained models in AES and discovered that BERT outperformed all of them achieving a QWK of 0.83.

2.1.1 Transformers

Transformer models work by using an attention mechanism that relates each token in the text to every other token thus enabling them to capture long-range contextual dependencies. According to [3], they make use of huge datasets of existing general text

data, such as Wikipedia Corpus, to pretrain multilayer neural networks with context-sensitive meaning of and relationships between words.

[4] compared the performance of several models on an AES task. The results show transformer models outperforming traditional models, achieving higher QWK scores. The best performer was a NCPR model, which used contrastive learning and utilised BERT for feature extraction achieved a QWK of 0.817, indicating a close similarity to human scoring.

[3] suggests not relying solely on transformer models and like [4], proposed a technique of combining transformers with more traditional approaches like those that employ handcrafted features. [1] argued that human evaluation remains important since expert markers can identify subtle linguistic and cognitive qualities that automated systems may overlook. [4] indicated that when models work in combination the results are more promising. [7] used a hybrid model – LSTM and Word Embedding and achieved a QWK of 0.94. [3], [4] and [1] all give compelling reasons for a hybrid design for an AES system but do not give explicit indication of the format that the essay must be in for evaluation.

Limitations: As effective as stand-alone neural approaches are, they are usually criticized for their lack of readability and interpretability by humans, as was mentioned in [5]. Several studies cited in [4] have shown that to be effective at AES tasks, transformer models require fine tuning.

2.1.2 Optical Character Recognition (OCR)

[8] defines OCR as a computational technology that extracts text in an editable format from PDF files, scanned or hand-written documents, images etc. [9] describes traditional OCR systems as comprising two modules: a text detection module and a text recognition module. Text detection localizes all text blocks at word-level or text-line level while text recognition aims to understand the text image content and transcribe the visual signals into natural language tokens [9].

Mostafa et al. [10] summarized work done using OCR to digitize Bangla scripts - a diverse and complicated language in written form. They concluded that CNN architectures are the most accurate tried and tested method for that task. Although no

evidence is given of the performance of transformer models, the researchers do acknowledge their potential. The decision to experiment with Bangla script as compared to English may be due to its perceived difficulty compared to English although the research did not make that claim.

Limitations: No mention is made of other architectures or listing of benchmarks for a performance comparison. Also, the study does not give evidence of CNNs being trained on English language typed or cursive handwriting datasets.

Tr-OCR

[9] achieved state-of-the-art results on printed, handwritten and scene text image datasets with Tr-OCR, an end-to-end Transformer-based model. Tr-OCR achieved word-level precision, recall and F1 scores of 96.59, 96.57 and 96.58 respectively on the SROIE (Scanned Receipts OCR and Image Formation Extraction) dataset. It also achieved a state-of-the-art CER (Character Error Rate) of 2.89 on the IAM Handwriting Dataset, a dataset containing, according to [11], natural variety, cursive elements and mixed script samples of handwriting.

Limitations: According to [12], Tr-OCR is designed for single-line text detection and failed to extract text from full-page or paragraph-level images. In one experiment it achieved a low accuracy of around 50-60% in text extraction.

BrainPower relies on full page text extraction so to utilise Tr-OCR essays must be segmented into a series of sentence-images, fed to the model and then recombined. This begs the question; can the OCR function of BrainPower utilise Tr-OCR's proven handwriting recognition and PaddleOCR's robust text detection capabilities?

A study carried out by [13], where YOLOv10, an object detector, was integrated with Tr-OCR and PaddleOCR for reliable text detection, found that Tr-OCR provided the best overall solution for reliable, high-accuracy container ISO code recognition with a recognition accuracy of 92% outperforming PaddleOCR, a hybrid CNN-Transformer model.

Limitations: The research was carried out on text printed/painted on containers. Such text is usually printed using a stencil ensuring uniform height and spacing. There is also

no evidence from this research to support the usage of YOLO and PaddleOCR in combination for detecting handwritten text.

PaddleOCR

[14] PaddleOCR is an open-source OCR model launched by Baidu Paddle. The architecture always maintains the three-stage pipeline design of "text-detection + direction classification + text recognition". [14] tested the model on a dataset containing 200 national grids stand slogans and power supply line maps collected on the spot at a power generation environment. The images were captured at different angles under varying lighting conditions. The model achieved high detection accuracy of over 95% for standardized slogans and power supply circuit diagrams. It also achieved 93.3% and 86.6% average recognition for standardized slogans and power supply circuit diagrams respectively [14].

Limitation: This study focused solely on block printed text, which is generally uniform, evenly spaced and aligned. As such there is no evidence to suggest that PaddleOCR can recognize handwritten text. However, it recognizes text from whole page documents, unlike Tr-OCR. These studies focused on OCR, but ASR presents an optional modality for presenting an essay for evaluation.

2.1.3 Automated Speech Recognition (ASR), Speech-To-Text (STT)

[15] A speech-to-text (STT) system accepts an audio input signal from either a person or a machine and detects and converts individual words and phrases into text. [16]

Advances in deep learning technology has profoundly transformed automatic speech recognition (ASR) over the past decade. Historically, ASR systems were constructed from a complex pipeline of independently trained components for example, a pronunciation model (lexicon) and a language model (LM). Like early AES systems, features had to be hand engineered, which was tedious and did not scale well.

[17] described how three major research groups - Microsoft Research (MSR), Google and IBM Research, achieved significant improvements in a variety of state-of-the-art ASR systems with Deep Neural Networks. According to [16], the modern era of ASR is defined by firstly, a shift towards end-to-end (E2E) models – systems that learn a direct

mapping from acoustic features to text using a single, jointly optimized neural network which simplifies training and improves performance. Secondly, a shift in the data paradigm. Early deep learning systems used supervised learning to train on large datasets of labelled data. Modern systems have moved towards unlabelled data. Self-supervised systems have demonstrated the ability to learn from raw audio data while weakly supervised systems have demonstrated the ability to generalise to data they were never trained on [16].

Whisper

[18] One model that exemplifies this shift is OpenAI's Whisper, a transformer. [19] For translation tasks, the Transformer can be trained significantly faster than architectures based on recurrent or convolutional layers. The Whisper model was trained on an enormous dataset of 680,000 hours of labelled audio. Whisper transcribed 950 out of 1000 words correctly in research carried out by [20] achieving 95% accuracy. The model's integration into classroom technology effectively supports students with diverse linguistic, cultural, and ability-based needs, achieving high accuracy and frequency responsiveness. Whisper is a valuable tool for educators seeking to foster a more accessible learning environment [20].

Limitations: Occasionally Whisper hallucinates. [20] discovered that about 1% of transcriptions contain fabricated phrases that were not present in the original audio, and that a significant portion of these can lead to harmful implications. The study further indicated that systems which lack the necessary processing power will experience transcription latency, as Whisper models occupy gigabytes of storage space and this may be undesirable.

2.1.4 Multimodal Systems

This section discusses some multimodal systems that share some similarity with BrainPower but however, fail to provide in entirety for all input modalities, catering to students who have writing disabilities. These observations present the following two gaps that this research hopes to fill:

Gap 1 - most AES systems ignore input equity.

Gap 2 – the architecture of commercial multimodal products is not accessible for analysis.

An extensive search of Google Scholar, IEEE Explore, ACM and similar services and querying current AI applications like ChatGPT and Gemini with the search terms "multimodal AES", "accessible multimodal AES" and "hybrid multimodal AES" reveals that such systems are exceptionally rare. The results returned were of systems that orchestrated at most two AI models of disparate modality. This indicates a gap in current AES technology revealing that most AES systems ignore input equity.

The architecture of commercially available multimodal systems such as IntelliMetric® are not freely available for analysis. My research project intends to fill the gap by implementing an accessible, interpretable, hybrid, multimodal AES system by orchestrating three pre-trained models – an OCR, ASR and a transformer model that accepts as input, written, typed and spoken essays and transparently scores them.

CAFES

[21] proposed a multimodal AES system called CAFES that uses a MLLM to process graphics e.g. a scenery along with text. The system orchestrates GPT-4o as the MLLM. The dataset used for their experiment is the ESSAYJUDGE dataset which consists of 1,054 multimodal essays written at the university level. This dataset is limited in that it does not cover complex visual inputs such as film frames.

Limitation(s): This system although multimodal, is still not designed to cater to students with disabilities, as it, like other AES systems ignores input equity.

ExaGrade

[22] developed a multimodal system ExaGrade that integrates optical character recognition (OCR) with the generative pre-trained transformer (GPT). The platform integrates two pre-trained models - HandwritingOCR to extract text from handwritten exam papers and like [21] employs GPT-4 to evaluate and score student responses. The platform offers features such as exam creation where for MCQs and TFs the teacher supplies the answers, and student exam completion using handwritten or online modalities.

Limitation(s): The platform is described as accessible; however, accessibility only encompasses availability online at GitHub since the platform is released as an open-source project [22]. ExaGrade is not accessible to students with writing disabilities. Notably absent from the system is a means to input an essay in audible form.

[23] proposed VisualAES, a bi-modal system that utilizes both textual and visual features for handwritten essay scoring. It orchestrates three pretrained transformer-based models to take both textual features and visual features extracted by Faster R-CNN and then uses a stacking ensemble model to robustly map their output to a final score. This system's architecture is more congruent with what is being proposed in this research – an OCR model to convert text from scanned essays to digital text for processing, and a transformer model for essay scoring. VisualAES was evaluated on the Automated Student Assessment Prize (ASAP) dataset and their own handwritten Chinese Student's handwritten essay dataset (ChnStd). Their results showed that their system outperformed all state-of-the-art methods on both datasets.

Limitation(s): VisualAES like CAFES does not provide accessibility for students suffering writing disability.

[20] developed an inclusive learning classroom using the OpenAI Whisper model. The model was selected to promote diversity and inclusivity in education through real-time transcription and translation for students with diverse linguistic and cultural abilities and backgrounds. The results were encouraging as the model achieved high accuracy especially in an environment where ambient noise did not alter the audio signal input. This system however is limited in input modality as it only focuses on audio transcription with no allocation made for submission of handwritten work.

[24] developed a system for investigating the possibility of automatically grading handwritten essays by leveraging OCR together with a fine-tuned and trained Large Language Model (LLM) Mistral 7b. According to the BERT Score analysis, the precision, recall and F1 scores were observed to be consistent over all the essays. The F1 score of 85.78% demonstrates the model's overall stability in replicating human-grading decisions with a high degree of consistency. The system used a Moodle plugin that integrated Mistral 7b into the platform.

Limitation(s): There is no mention of inclusivity in the system developed by [24] showing once again that most modern AES systems focus on the quality of essay scoring and do not prioritize input equity, emphasizing the need for an integrated system.

2.2.0 Challenges and consideration for component selection

The datasets

According to [3], the most elaborate AES systems currently have trouble accurately evaluating the coherence and persuasiveness of an essay, or an argument's clarity, and that is due to two main reasons: (1) such aspects are very hard to model, and (2) the lack of datasets that provide human-graded dimension-specific data, especially compared to those that provide holistic scoring. [5] indicates that the problem originates from AES research relying heavily on the ASAP dataset as the primary benchmark, highlighting the limited options available. Additionally, most publicly available datasets are tailored toward English-language assessment e.g., TOEFL11, CELA, or the Cambridge English assessment datasets.

[21] further emphasizes this limitation, describing the ASAP and its extended version, ASAP++, as both containing few topics. ASAP++ adds trait-level annotations but merges key content traits into a single "CONTENT" label, decreasing granularity and interpretability. The TOEFL11 dataset, similarly, uses only coarse-grained proficiency labels (low / medium / high). Thus, models trained on the TOEFL11 dataset will inherently produce results that are difficult to interpret due to the granularity of the training data [21].

[5] also points out that the dataset availability and diversity issue is not limited to the field of research, but extends to commercial products, which primarily focus on English-language datasets. The quality, quantity, and availability of the data are fundamental to the model's performance. It is fitting to consider available models, discuss the impact the limited datasets have on them and propose the best option to provide reliable, interpretable trait scoring.

The models

[5] presents some of the issues with Transformer pretrained models as having a closed-box nature, high computational requirements, longer training times and the need for large datasets. Additionally, issues presented by [5] are data quality, biases in training data, generalization, and overfitting which still pose significant obstacles that need to be addressed.

[22] cited three studies that showed the limitations of ChatGPT, a transformer model, as an AES tool. The results indicated that ChatGPT often scores lower than human scorers and its evaluation is inconsistent due to variations in its scoring mechanism, training data, model updates, and ingrained randomness. [22] Additionally, the consistency of ChatGPT's scores over multiple evaluations was low, indicating variability and raising concerns about its reliability.

[25] explored the potential of using GPT-based models as an alternative solution to the data scarcity issue, comparing their performance to that of fine-tuning BERT models, which remains the state-of-the-art, in quantitative text analysis. They concluded that BERT models remain the overall go-to method for classification although GPT models can serve as alternatives on smaller tasks. GPT models are also easier to tweak; however, they are more expensive. Due to the findings of [22] and [25], GPT will not be included in this system.

[26] DeBERTa (Decoding-enhanced BERT with Disentangled Attention) is an improvement of previous state-of-the-art pre-trained transformer models such as BERT and RoBERTa, its predecessors. [26] BERT and RoBERTa represent each word in the input layer by a single vector which is the sum of its word (content) embedding and position embedding. Each word in DeBERTa, however, represents a word with two vectors that encode its content and position, respectively, and the attention weights among words are computed using disentangled matrices based on their contents and relative positions [26].

[26] DeBERTa, after training on 78GB of data, outperformed RoBERTa, which was trained on 160GB of data and surpassed human scoring on the SuperGLUE benchmarks for the

first time since 2021. This benchmark includes a wide range of tasks such as question answering, natural language inference, and word sense disambiguation [26].

DeBERTa has been selected to form the core component in the essay processing pipeline for BrainPower, serving as the final step before the end user receives a breakdown of how the traits are scored. This decision is based on the empirical research that showed it outperforming human scorers and other state-of-the-art models on established benchmarks, making it more suitable for the AES task. DeBERTa is also freely available from the Hugging Face platform and offered under the MIT licence, facilitating freedom of modification and reproducibility.

Limitations of OCR in general

The accuracy of OCR models in recognizing text from images depends on the quality of its input image. High-quality images have clear, uncluttered text, no background noise (such as smudges) and are captured in well-lit environments. Most classroom environments, however, do not provide all these conditions. Furthermore, quality of images is determined by the photographer's ability to hold the camera steady and the quality of the camera used to capture the image. Consequently, selection of an appropriate model is critical to preventing errors from propagating through the pipeline as errors fed into the core (DeBERTa) will degrade its scoring accuracy.

Several popular models struggle to decipher handwritten text or are only capable of extracting text from one language. Others are limited to handling only single lines of text and cannot scan whole documents, making it necessary for a text extraction add-on. One such example is Tr-OCR, according to [12], which is designed for single-line text detection and failed to extract text from full-page or paragraph-level images, in one experiment achieving a low accuracy of around 50-60% in text extraction [12].

It is for these reasons, along with the findings of [14] discussed previously, that PaddleOCR has been selected as the OCR input component for BrainPower. Since it is beyond the scope of this research to fine-tune a model, the capabilities of PaddleOCR make it an ideal candidate for inclusion in BrainPower's AES pipeline.

Limitations of ASR (STT) in general

[27] noise interference, echoes, accents, dialect variation and overlapping speech have a degrading effect on the performance of speech-to-text (STT) and text-to-speech (TTS) systems, resulting in transcription errors. As with OCR systems, ASR systems are vulnerable to environmental conditions as these determine the quality of the input, which invariably determines the quality of the output. According to [27], model and dataset scarcity are the reasons why the issues mentioned earlier still pose a challenge to modern ASR systems.

This issue of dataset scarcity seems to be a common thread among modern AI models, since they rely heavily on data for training and testing, underscoring the need for greater diversity and availability. These limitations are taken into consideration however, and Whisper will serve as the ASR/STT component of BrainPower due to its accessibility and state-of-the-art performance. The results from both component and integration testing will indicate if the model needs to be fine-tuned or replaced.

Limitations of Multimodal Integrated Systems

Multimodal integrated systems inherently suffer the same issues as their constituent components – OCR, ASR, and Transformer models. Additionally, they face issues such as:

- Computational complexity: orchestrating three pretrained models requires a lot of processing power and storage space.
- Error propagation: It is likely that mistakes from either the audio transcription done by the ASR or from recognition and digitization by the OCR will flow as input to the AES system core. If errors in the input data are not detected and handled correctly, they will propagate and reduce the core component's accuracy and reliability, resulting in inaccurate essay scores.
- Modality imbalance: Typed essays may score higher than handwritten or audio-transcribed essays due to noise introduced by OCR or the accent or dialect of the speaker for example.

- Data Bias: The pre-trained models are inherently biased towards the data they were trained on.

It is therefore incumbent on the designer to mitigate error propagation by detecting errors in the output from the ASR and OCR using metrics such as WER and CER and managing them at the component level, alerting human scorers when review is necessary.

Chapter 3

Design

Project Management

Overleaf is a Gantt chart, displaying all tasks that comprise the workflow for the development of system, and the estimated time required for their completion. This chart does not indicate the presence of dependencies, however, these will be outlined in this section. Some of the key milestones and dependencies are as follows:

- **Component Tests (ASR, OCR, Transformer):** These milestones are critical because the ASR, OCR and Transformer models are the three main components of the AES system – failure in either the ASR or OCR component will propagate and results in an inaccurate, unreliable essay score. Successful results here move the system towards completion.
- **Sample collection:** The collection of samples (images and audio of essays), is a key milestone because it allows the system to be tested on a component level and as a complete unit.
- **Integration Testing:** completion of this task is critical as the User Interface Design process requires the system's output to provide feedback to the end-user from the AES evaluation process. Success also indicates that the penultimate hurdle has been achieved and the system is near-ready for presentation to the end-user.
- **User Interface Design:** this task is critical because **user testing** depends on its completion. A fully functional interface is necessary for the user to interact with the system and provide timely feedback that will be used to make improvements to the system.
- **User Testing:** This stage is critical because success here is the sole metric for evaluating the system and determining if the end-goal has been achieved.

Gantt Chart

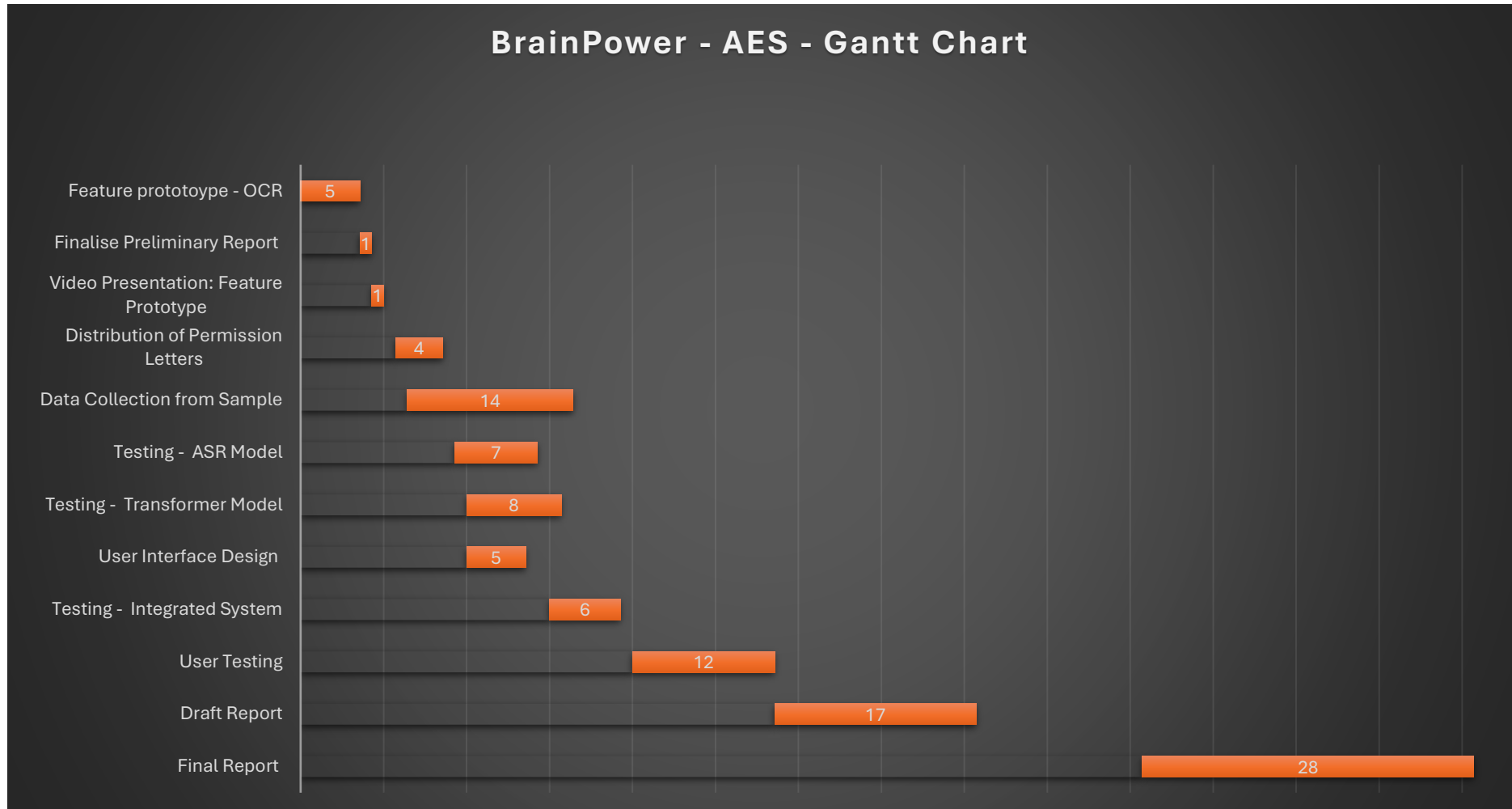


Figure 1 Gantt Chart for BrainPower AES project

Design

The diagram below shows a high-level design of the BrainPower system. The system will contain three (3) core components – two (2) input components, ASR and OCR. The ASR model, (OpenAi Whisper) will transcript spoken essays to digital text. The OCR model, (Paddle-OCR), which detects and recognises text in a JPEG image and records this as digital text. The core of the system is the transformer (DeBERTa) that accepts input essays in digital text format from the other two components and scores the input essay according to a rubric.

BrainPower (multi-modal AES) PIPELINE

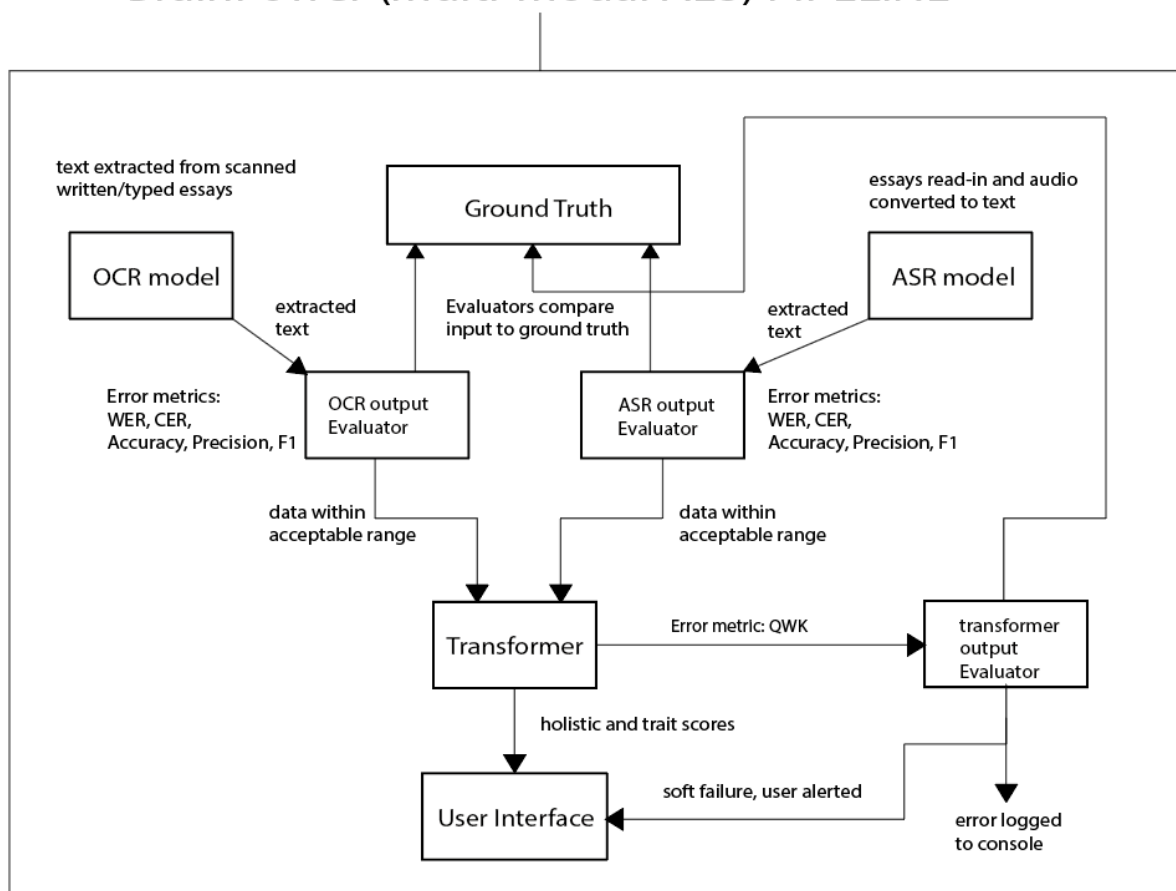


Figure 2 Architecture of BrainPower

The system will use the OCR and ASR evaluation logic to filter essays based on the WER and CER forwarding them to the transformer if they are within the acceptable threshold and to the human scorer if they are not.

Typical Workflow:

Scenario 1:

- Input (image of handwritten or typed essay) submitted to the OCR model
 - Model extracts text from the image and outputs digital text to its Evaluator
 - OCR Evaluator runs metrics such as WER, CER etc. referencing ground truth
 - Results within acceptable range: text forwarded to transformer for evaluation
 - Unacceptable results: system displays message to user, fails softly, logs details

Scenario 2:

- Input (read-in essay) submitted to the AST/STT model
 - Model extracts text from the audio and outputs digital text to its Evaluator
 - ASR Evaluator runs metrics such as WER, CER etc. referencing ground truth
 - Results within acceptable range: text forwarded to transformer for evaluation
 - Unacceptable results: system displays message to user, fails softly, logs details

Scenario 3:

- Input (essay in text format) submitted to the Transformer model
 - Model processes text and logs results to its Evaluator
 - Transformer Evaluation runs metrics such as QWK referencing ground truth
 - Results within acceptable range: text displayed to human scorer
 - Unacceptable results: system displays message to user, fails softly, logs details

Chapter 4

Implementation

Procedure

To evaluate the performance of the system the following approach will be used:

Data input to OCR – WER and CER metrics will be calculated and used to determine whether the output should be sent to the core for further processing or a flag raised for human scorer intervention

Data input to ASR- WER and CER metrics will be calculated and used to determine whether the output should be sent to the core for further processing or a flag raised for human scorer intervention

Data input to AES – will output trait scores along with a QWK which will be used to describe proximity of the system's scores to that of human scorers along

Datasets –this research will use the ASAP 1.0 dataset for training and testing
BrainPower

Testing Plan

About 25-30 students will be selected to form the sample. Each student will produce one handwritten essay, after which they will snap a photograph with their phone and forward for collection. The students will also be required to record a reading of the essay. Thus the sample data should comprise at least 25 written and 25 audio samples of the same essay. Only one prompt will be used, the purpose of this test is to determine whether the ocr and asr are suitable for inclusion in the AES system.

What model? What parameters? System specs model will be run on?

Reached out to 42 students in the age range of 12 – 21 (use lit to support sample size). Part of the reason for such a number is realistically all will not respond and a sample of 30 student is ideal as it is roughly the average population of a class. Used that age range since the original essay was written by high school students. Used random sampling, which was purposive due the targeted age group.

System Deployment/Implementation Details

Scripts to download models

Scripts to upload data for processing

Installation instruction, requirements.txt installation

The system will be implemented using Jupyter Notebooks. This resource contains sufficient facility to demonstrate the utility of the AES system. It is envisaged the code can be easily adapted for a Django implementation.

CHAPTER 5

Evaluation

5.1 Test Environment

GitHub Repo URL - <https://github.com/AsaDorville758/aes-thesis>

Operating System: Windows 11 Professional

IDE: Jupyter Notebook

Python: 3.11.9

Hardware:

- **Processor:** AMD Ryzen 7 5700G 8-Core
- **RAM:** CORSAIR Vengeance LPX DDR4 32GB
- **GPU:** ZER-LON GeForce GTX 1660 Super 6GB GDDR6 192Bit PCIE 3.0X16

Dataset: ASAP 1.0

OCR Model: PaddleOCR PP-OCRv6

ASR Model: Whisper Large v3

Transformer Model: QWEN

Django: 5.2.16

Database: SQLite3

5.2 Unit Tests

1. Extract and display the reference essay – Essay 1 of Prompt 1

Test environment: Jupyter Notebook

Test	Description	Expected Results	Actual Results
UT001	Import dataset	Dataset loads successfully	Dataset loaded successfully
UT002	Extract and display Prompt1(see Appendix A) reference essay from ground truth the ground truth dataset	Reference essay successfully extracted from dataset and displayed	Reference essay successfully extracted from dataset and displayed

```
print(reference_essay)
```

Dear local newspaper, I think effects computers have on people are gre
ple, helps us learn about the globe(astronomy) and keeps us out of tro
s on the phone with friends! Do you ever time to chat with your friend
uter, theirs plenty of sites on the internet to do so: @ORGANIZATION1,

2. Calculate and display word and sentence count from the reference essay

Test	Description	Expected Results	Actual Results
UT003	Display Word Count	Number of words: 331	Number of words: 331
UT004	Display Sentence Count	Number of sentences: 16	Number of sentences: 16

```
#Display Extracted Features - Word and Sentence Counts  
extract_features(reference_essay)
```

Number of words: 331
Number of sentences : 16

The natural language tool kit was used to tokenise sentences and words to provide a count that will later be used as a point of reference in determining the accuracy of the OCR detection and ASR transcription processes.

Purpose: Although these features will not contribute to the scoring of the essay by the transformer, they serve to satisfy the requirements for the system to be interpretable and explainable to human scorers.

3. Calculate and display the word and sentence error rates

Test	Description	Expected Results	Actual Results
UT005	Calculate and display the word error rate (WER)	Word Error Rate: 0.018	Word Error Rate: 0.018
UT006	Calculate and display the character error rate (CER)	Character Error Rate: 0.046	Character Error Rate: 0.046

A copy of the reference essay was deliberately modified by appended the digit “5” to the first three and last three words of the first and last sentences. The ‘wer’ and ‘cer’ methods from the **jiwer** module were then used to calculate the word and character error rates.

Purpose: This was designed to test the accuracy of the `calculate_error_rates()` function, prior to the implementation of the OCR and ASR components. These metrics are essential to the operation of the evaluator functions in determining what output fed to the transformer for further processing or returned to the human scorer for manual scoring.

4. Normalize the reference essay

Test	Description	Expected Results	Actual Results
------	-------------	------------------	----------------

UT007	Convert a body of text to lower case	Essay successfully converted to lower case	Essay successfully converted to lower case
UT008	Normalise symbols	Symbols normalised successfully	Symbols normalised successfully
UT009	Normalise spoken forms	Spoken forms normalised successfully	Spoken forms normalised successfully
UT010	Clean text by removing all punctuation and extra spaces	All punctuation and extra spaces successfully removed from text	All punctuation and extra spaces successfully removed from text

For efficiency, scalability and modularity, a class based approach – the EssayEditor class, was used to encapsulate the functions that will be applied to an essay to normalise it.

```
#REGEX - compile patterns and rules once for efficiency
REPLACEMENTS = {
    '/': ' or ',
    'slash': 'or',
    '@caps1': 'at caps one',
    '@caps2': 'at caps one',
    '@date1': 'at date one',
    '@organization1': 'at organization one',
    '@organization2': 'at organization two',
    'globe(astronomy)': 'globe astronomy',
}
SYMBOL_PATTERN = "|".join(re.escape(key) for key in REPLACEMENTS.keys())
COMPILED_SYMBOL_PATTERN = re.compile(SYMBOL_PATTERN)
PUNCTUATION_RULE = re.compile(r"^[^\s\w]")#find all characters that are not
SPACES_RULE = re.compile(r"\s+")#find all occurrences of multiple spaces
```

The researcher chose to use class attributes instead of instance variables for the normalisation process to conserve memory – each object will have its own copy of instance variables as compared to one set per class, available to all objects. This means that the normalisation process scales well as the number of input essays grows.

Some members of the sample use the spoken form slash to represent ‘/’, while others use ‘or’. The normalise () method of the EssayEditor class, substitutes these symbols

with their spoken forms. This is to ensure that when applied to spoken essays from different speakers, the output is consistent after normalisation, making calculation of the CER and WER more accurate.

```
nt right you might not know where your child is at caps one forbid  
ning chatting or just playing games safe and sound in your home or  
because computers can have great effects on you or child because i  
believe or not keeps us out of troble thank you for listening
```

Normalised:

1: Essay converted to lower case successfully.

2: Symbols and spoken forms normalized successfully.

3: Cleaned- punctuation an unwanted spaces removed successfully.

The screenshot above shows the output from the EssayEditor object its text – the reference essay, has been normalised. The object also prints confirmation of all successfully executed operations – when its show_edit_log() method is invoked.

5.3 Component Testing

Note: Most tests on the batch of handwritten essays for OCR evaluation, have been completed, however, the evaluation analysis is still outstanding. Also, experimentation to determine the best configuration to achieve the lowest wer and cer are also still outstanding.

See appendix B for Jupyter Notebook code if necessary or visit GitHub repo, see URL under test environment section.

1. OCR: paddleOCR

- **Recognition model:** PP-OCRv6_medium_det
- **Detection model:** PP-OCRv6_medium_rec

Test environment: Jupyter Notebook

Test	Description	Expected Results	Actual Results
CT001	Load paddleOCR model	Model loaded successfully.	Model loaded successfully.
CT002	Load handwritten essay image	Image loaded successfully	Image loaded successfully
CT003	Detect text regions	Text regions successfully detected	Text regions successfully detected
CT004	Recognize handwritten text	Handwritten text successfully recognised	Handwritten text successfully recognised
CT005	Calculate WER and CER	WER and CER successfully calculated	WER and CER successfully calculated
CT006	Display OCR evaluation	Evaluation results successfully displayed	Evaluation results successfully displayed

2. ASR: OpenAi Whisper

- **Models:** faster-whisper-base, faster-whisper-small, faster-whisper-medium

Test environment: Jupyter Notebook

Test	Description	Expected Results	Actual Results
CT007	Access audio file(s) via specified path	Model loaded successfully.	Model loaded successfully.
CT008	Load audio file to librosa for signal conversion	Audio loaded successfully	Audio loaded successfully
CT009	Generate transcription for a single .wav file	Transcription performed successfully	Transcription performed successfully
CT010	Load whisper model(s)	Models loaded successfully	Model loaded successfully
CT011	Transcribe an audio recording	Recording successfully transcribed	Recording successfully transcribed
CT012	Save model configuration and experiment data.	Model configuration and experiment data successfully saved	Model configuration and experiment data successfully saved
CT013	Transcribe a batch of .wav spoken essay files	Batch of spoken essay files successfully transcribed	Batch of spoken essay files successfully transcribed
CT014	Assign a transcription “pass” or “fail” based on its WER and CER values	Transcriptions successfully assigned “pass” or “fail” status based	Transcriptions successfully assigned “pass” or “fail” status based

		on WER and CER values	on WER and CER values
CT015	Save the WER and CER statistics for the batch of recordings	WER and CER statistics per batch saved	WER and CER statistics per batch saved

Selecting the best model along with the optimum parameter configuration. A list of six, 2-tuples – (**model size, compute_type**) parameter configurations were used to select the best model for the system.

```
experiments= [] #stores experiment data as dictionaries
size_compute_type_pairs= [
    ('small', 'int8_float16'),
    ('small', 'float16'),
    ('base', 'int8_float16'),
    ('base', 'float16'),
    ('medium', 'int8_float16'),
    ('medium', 'float16'),
]
```

Thus, a total of 30 experiments – 5 per each **model size, compute_type** pair, were carried out. For each pair, transcription was done using beam sizes from 1 to 5 respectively. The results stored as a list of dictionaries are shown in the table below.

	Model Size	Compute Type	Beam Size	Model Load Time	Transcription Time	WER	CER
24	medium	int8_float16	5	4.02	14.99	0.089	0.049
28	medium	float16	4	2.33	51.76	0.089	0.049
22	medium	int8_float16	3	4.02	14.32	0.092	0.050
29	medium	float16	5	2.33	49.95	0.092	0.051
3	small	int8_float16	4	1.73	7.19	0.092	0.054

Experiments 24 and 28 achieved the same transcription accuracy. However, they differed noticeably in the time the model took to load. In experiment 24, mixed-precision quantization - int8_float16 was used, along with the **beamsize=5**. This model took roughly four (4) seconds to transcribe the audio.

The model in experiment 28, with **beam size=4** and **compute_type=float16**, however, took about 50% less time to load but was more than three times slower than the model in experiment 24.

(Questions requiring answers)

Why was the VAD Filter set to True?

Why was the beam size adjusted?

Why were 16 threads alone used?

Why was CUDA fixed?

After the optimum configuration was implemented, the **details** dictionary was updated to include not only each file's properties e.g. length and size along with the WER and CER for the transcription but also a "pass" or "fail" evaluation status. These values are assigned in relation to the WER and CER values exceeding the set threshold values of 0.099 and 0.059 respectively. These evaluation labels will be used later, to determine which transcriptions are forwarded to the transformer or human scorer for scoring.

A **batch_transcription** dictionary stores all individual transcription-detail dictionaries in a list along with WER, CER and evaluation statistics for the batch.

```

batch_transcription= {
    "details": [],
    "wer_stats": {
        "mean": '',
        "median": '',
        "min": '',
        "max": '',
        "std_dev": '',
    },
    "cer_stats": {
        "mean": '',
        "median": '',
        "min": '',
        "max": '',
        "std_dev": '',
    },
    "eval_stats":{
        "wer passes": 0,
        "wer fails": 0,
        "wer pass percentage": 0,
        "wer fail percentage": 0,
        "cer passes": 0,
        "cer fails": 0,
        "cer pass percentage":0,
        "cer fail percentage":0,
        "total passes": 0,
        "total fails": 0,
        "total pass percentage": 0,
        "total fail percentage": 0,
    }
}

```

The rationale behind this design decision is to make possible the analysis of transcription statistics per individual transcription and per batch (e.g. a class of 25 students).

Although the WER and CER per batch are stored in the **batch_transcription** object, their calculation is not performed locally. The global **calculate_stats** function is invoked to perform these calculations. **calculate_stats** converts a list of individual transcription statistics to a **numpy** array and invokes numpy's statistical functions to obtain an aggregate of the mean, median and so on. The function returns a dictionary with the calculated statistics. The rationale behind this design, is to avoid code redundancy – the OCR component also utilises the **calculate_stats** function.

	WER	CER
mean	11.55%	5.47%
median	10.56%	4.70%
min	6.67%	1.92%
max	21.94%	13.13%
std_dev	3.76%	2.82%

Evaluation of Whisper:

The mean WER, 11.55%, is slightly higher than the acceptable threshold of 9.0%. The max WER however, differs from the min by roughly 15 percentage points – indicating variation in transcription accuracy across the samples. The median WER, 10.56% is slightly above the threshold while the CER, 4.70%, is slightly below the threshold. Whisper transcribed 27 essays. 29.6% were within the acceptable bound and were classified as “pass”, which means that they will be scored by the transformer. The majority, however, more than double the percentage pass, were classified as “fail”, and will be assigned to the human scorer for scoring.

The number of essays classified as “fail”, due to a high WER is more than double that of those classified as “pass”. With respect CER the number of transcriptions that are classified as “pass” is roughly 10% more than those classified as “fail”. The model seems to more consistently get words wrong rather than individual letters or characters. There were 8/27 WER passes, 29.63% and 15/25 CER passes, 55.56%.

The most accurately transcribed essay has a WER of 6.67%, and a CER of less than 2%. Perusal of this essay after normalisation revealed that it is near perfect and that the errors in transcription did not detract from the semantic of the essay. An excerpt is shown below.

Student ID: Neilage

WER: 6.67%

CER: 2.45%

Evaluation Status: pass

dear local newspaper i think apex computers have on people helps us learn about the globe astronomy and keep on the phone with friends do you ever tend to chat with heres plenty of sites on the internet to do so at organizing up meeting with your boss on the computer your teenagers learn about other countries or dates outside of yours well think your child spends a lot of time on the computer

The essay with the median WER percentage has a WER slightly above the acceptable WER threshold, and a CER slightly below the acceptable threshold.

Student ID: AnonTwo

WER: 10.56%

CER: 3.58%

Evaluation Status: fail

dear local newspaper i think effects computers have on people are grow people help us learn about the globe astronomy and keep us out of days on the phone with friends do you ever have time to chat with your computer theres plenty of sites on the internet to do so at organization youre setting up meetings with your boss on the computer your teenagers did you learn about other countries or states outside of yours well time you might think your child spends a lot of time on the computer

From a visual inspection of the normalised essay, it is apparent that most of the wrongly transcribed words are still decipherable by a human scorer. For example, Whisper transcribed “sea floor spreading” as

the economy see flow spreading

. This observation, however, does not motivate an adjustment to the current thresholds as this substitution, though recognisable to a human scorer, is likely to affect the transformer’s interpretation, resulting in an inaccurate score.

The screenshots below are from transcriptions with WER and CER percentages between the min and median and the median and max, respectively.

Student ID: Daniella

WER: 8.89%

CER: 3.79%

Evaluation Status: pass

dear local newspaper i think effects computers have on people are great
le helps us learn about the globe astronomy and keeps us out of trouble
the phone with friends do you ever have time to chat with your friends
theres plenty of sites on the internet to do so organization one at org
g up meeting with your boss on the computer your teenager is having fun
n about other countries states outside of yours well i have my computer

Student ID: Hannah

WER: 13.06%

CER: 5.39%

Evaluation Status: fail

dear local newspaper i think effects computers have on people are great lea
w people helps us learn about the globe astronomy and keeps us out of troubl
lways on the phone with friends do you ever have time to chat with your fri
computer theres plenty of sites on the internet to do so at organization onl
le youre setting up a meeting with your boss on the computer your teenager :
how did you learn about other countries or states outside of yours well i h
in our time you might think your child spends a lot of time on the computer

The transcription of Daniella voice is highly accurate with a WER of 8.89% and a CER of 3.79%. The transcription of Hannah although classified as “fail” is still readable and interpretable by a human scorer.

The most inaccurate transcription is still decipherable by a human scorer, as seen in the screenshot below.

Student ID: Gordon

WER: 21.94%

CER: 13.13%

Evaluation Status: fail

they are local newspapers i think if you are from these foreign
riends or new people helps us learn about the global astronomy
ja is always on the phone with friends do you ever have time t
at no computer they spend their time on the internet to do so
g with your boss on the computer your teenager is having fun o
er countries or dates outside of yours well i have my computer

This indicates that the evaluation logic worked as intended, demonstrating sensitivity to transcriptions with high WER percentages, preventing them from moving through the pipeline.

This analysis supports the inclusion of Whisper in the AES pipeline as an assistive technology. However, the high proportion of failed transcriptions justifies the use of

evaluation logic to route transcriptions to the appropriate scorer based on their WER and CER.

Transformer: DeBERTa

- **Model:**

Test Environment: Jupyter Notebook

Test	Description	Expected Results	Actual Results
CT012	Load model	The model loaded successfully	
CT013	Construct evaluation prompt	The evaluation prompt was successfully constructed	
CT014	Submit essay for scoring	The essay was successfully submitted for scoring	
CT015	Generate holistic score	A holistic score was generated successfully	
CT018	Generate trait scores (if possible)	Trait scores were successfully generated	
CT016	Generate feedback	Feedback was successfully generated	
CT017	Produce structured output (JSON or formatted response)	Output was structured successfully as JSON	

5.4 Integration Testing

Test	Description	Expected Results	Actual Results
IT001	Verify that OCR output is successfully passed to the transformer	The output was successfully passed to the transformer	
IT002	Verify that OCR evaluation metrics is successfully stored in the database	The output was successfully stored in the database	
IT003	Verify that ASR output is successfully passed to the transformer	The output was successfully passed to the transformer	
IT004	Verify that ASR evaluation metrics is successfully stored in the database	The output was successfully stored in the database	
IT005	Verify that the output from the transformer is successfully stored in the database	The output was successfully stored in the database	
IT006	Verify that the output from the transformer is successfully stored displayed to the user interface	The output was successfully displayed to the user interface	

5.5 Functional System Testing

These tests are designed to evaluate the systems performance under real-world conditions.

Test	Description	Expected Results	Actual Results
FST001	System accepts an image of written essay, paddleOCR extracts text, OCR Evaluator determines the text recognition accuracy to be acceptable and forwards essay to DeBERTa	Trait and holistic scores are displayed via the interface	
FST002	System accepts an audio file of a spoken essay, Whisper transcribes audio, ASR Evaluator determines transcription accuracy to be acceptable and forwards essay to DeBERTa	Trait and holistic scores are displayed via the interface	
FST003	System accepts an image of written essay, paddleOCR extracts text, OCR Evaluator determines the text recognition accuracy to be unacceptable	User is alerted of the evaluation and directed to mark the essay manually	
FST004	System accepts an audio file of a spoken essay, Whisper transcribes audio, ASR Evaluator determines	User is alerted of the evaluation and directed to mark the essay manually	

	transcription accuracy to be unacceptable		
FST005	The system calculates the QWK after the transformer scores and essay	The system produced a QWK after the transformer scored the essay	

5.6 End-User Testing

Test	Description	Expected Results	Actual Results
EUT001	User can upload image of written essay	User successfully uploaded an image of a handwritten essay	
EUT002	User can upload audio recording of spoken essay	User successfully uploaded an audio file of a spoken essay	
EUT003	User can interpret trait and feature scores	User successfully interpreted feature scored	
EUT004	User can interpret holistic score	User successfully interpreted holistic score	

5.7 Usability Testing

Statement	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
The system is easy to use					
The feedback is easy to understand					
The scoring feedback is useful					
The scoring feedback is verbose					
The trait scores are reasonable					
The holistic score is reasonable					
The interface is well organised					
The font size is appropriate					
The colour scheme is clear					
I would use this system again					

Chapter 6

Conclusion and Future Improvements

Future improvements:

- The teacher should be able to mark an essay rejected by the OCR or ASR evaluators, via the user interface and save the changes to the database.
- The teacher should be shown the essay with tokens highlighted/selected (like easyOCR check notebook, include a screen shot if necessary), to demonstrate the detection capability of the OCR detection model
- The system should provide a list of all words identified by the recognition model along with their confidence scores.
- The system should allow a human scorer to assign confidence scored to words to improve the model's recognition capabilities.
- The system should allow a human scorer to flag any words that are false positives. This data can be stored in a dictionary. This data can be used to train the model later.
- The system should be able to inform the student of the issues with the submission e.g. too dark or wrong angle or highlight words that have low confidence scores.
- The system should give suggestions for improving submissions e.g.
 - A word or series of words is illegible and requires rewriting, the system can highlight the objects and then suggest that the student rewrite more legibly for example in block letters or select letters from a software keyboard to identify the illegible letters.
- File conversion – accepting several different file formats to improve usability
- Combining interpretable features such as word count, sentence count, spelling errors with transformer output, or using them to engineer a trainable system.
- Merge images of essays submitted as separate files

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Appendix A - Consent Forms

Participant Consent Form (PR/001):

Project Title: BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system

Declaration by Participant

Please tick to confirm:

☒ I have read the **Participant Information Sheet (PR/002)**, overleaf, and consent to take part in **BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system** by submitting both an image of a essay written in my handwriting and an audio recording of my voice reading that same essay.

If you are happy to be identified in materials (see PR/002) relating to you participation, please tick each of the options that you agree to:

- ☒ I consent to being **referred to by name**
- ☒ I consent to being **referred to by my place of work**
- ☒ I consent to being **referred to by my position / title**

If you **wish to be fully anonymised** in any such materials, please instead tick here only: ☐

Name of Participant (please print): Aaliyah Monroe

Contact details of the Participant: Email monroseaaliyah@gmail.com Phone 17587256296

Date (DD/MM/YYYY): 06/08/2026

Declaration by Researcher

I have provided the above participant with "Participant Information Sheet (PR/002)", given the participant the opportunity to ask further questions, and believe that the participant has understood the process.

Name of the Researcher (please print): **Asa Dorville**

Date (DD/MM/YYYY): **23/07/2026**

Participant Consent Form (PR/001):

Project Title: BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system

Declaration by Participant

Please tick to confirm:

☒ I have read the **Participant Information Sheet** (PR/002), overleaf, and consent to take part in **BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system** by submitting both an image of a essay written in my handwriting and an audio recording of my voice reading that same essay.

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- ☒ I consent to being **referred to by my place of work**
- ☒ I consent to being **referred to by my position / title**

If you **wish to be fully anonymised** in any such materials, please instead tick here only: ☐

Name of Participant (please print): Alliah. C.A. Florius

Contact details of the Participant: Email floruscalleh1@gmail.com Phone (753) 713-7331

Date (DD/MM/YYYY): 29/01/2006

Declaration by Researcher

I have provided the above participant with "Participant Information Sheet (PR/002)", given the participant the opportunity to ask further questions, and believe that the participant has understood the process.

Name of the Researcher (please print): **Asa Dorville**

Date (DD/MM/YYYY): **23/07/2026**

Participant Consent Form (PR/001):

Project Title: BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system

Declaration by Participant

Please tick to confirm:

☒ I have read the **Participant Information Sheet (PR/002)**, overleaf, and consent to take part in **BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system** by submitting both an image of a essay written in my handwriting and an audio recording of my voice reading that same essay.

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- ☐ I consent to being referred to by my place of work
- ☐ I consent to being referred to by my position / title

If you **wish to be fully anonymised** in any such materials, please instead tick here only: ☐

Name of Participant (please print): _____ Bevan Jeria _____

Contact details of the Participant: Email _____ Phone __489-0721_____

Date (DD/MM/YYYY): _____ 5/08/2026 _____

Declaration by Researcher

I have provided the above participant with "Participant Information Sheet (PR/002)", given the participant the opportunity to ask further questions, and believe that the participant has understood the process.

Name of the Researcher (please print): **Asa Dorville**

Date (DD/MM/YYYY): **23/07/2026**

Participant Consent Form (PR/001):

Project Title: BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system

Declaration by Participant

Please tick to confirm:

☒ I have read the **Participant Information Sheet** (PR/002), overleaf, and consent to take part in **BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system** by submitting both an image of a essay written in my handwriting and an audio recording of my voice reading that same essay.

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- ☒ I consent to being **referred to by my place of work**
- ☒ I consent to being **referred to by my position / title**

If you **wish to be fully anonymised** in any such materials, please instead tick here only: ☐

Name of Participant (please print): Bryce Haughton

Contact details of the Participant: Email bryce.haughton@mdcs.edu.lc Phone 728-7226

Date (DD/MM/YYYY): 10-08-2026

Declaration by Researcher

I have provided the above participant with "Participant Information Sheet (PR/002)", given the participant the opportunity to ask further questions, and believe that the participant has understood the process.

Name of the Researcher (please print): **Asa Dorville**

Date (DD/MM/YYYY): **23/07/2026**

Participant Consent Form (PR/001):

Project Title: BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system

Declaration by Participant

Please tick to confirm:

☒ I have read the **Participant Information Sheet (PR/002)**, overleaf, and consent to take part in **BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system** by submitting both an image of a essay written in my handwriting and an audio recording of my voice reading that same essay.

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- ☒ I consent to being referred to by name
- ☒ I consent to being referred to by my place of work
- ☒ I consent to being referred to by my position / title

If you wish to be fully anonymised in any such materials, please instead tick here only: ☐

Name of Participant (please print): DANIELLA SYMPHORIEN

Contact details of the Participant: Email Symphorien.daniella@gmail Phone 384 3346

Date (DD/MM/YYYY): 13/04/2009

Declaration by Researcher

I have provided the above participant with "Participant Information Sheet (PR/002)", given the participant the opportunity to ask further questions, and believe that the participant has understood the process.

Name of the Researcher (please print): **Asa Dorville**

Date (DD/MM/YYYY): **23/07/2026**

Participant Consent Form (PR/001):

Project Title: BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system

Declaration by Participant

Please tick to confirm:

☒ I have read the **Participant Information Sheet (PR/002)**, overleaf, and consent to take part in **BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system** by submitting both an image of a essay written in my handwriting and an audio recording of my voice reading that same essay.

If you are happy to be identified in materials (see PR/002) relating to you participation, please tick each of the options that you agree to:

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- ☐ I consent to being **referred to by my place of work**
- ☒ I consent to being **referred to by my position / title**

If you **wish to be fully anonymised** in any such materials, please instead tick here only: ☐

Name of Participant (please print): Daniel Tsiboe

Contact details of the Participant: Email DanielTsiboe@gmail.com Phone +1 (256) 724-3002

Date (DD/MM/YYYY): 07/05/2026

Declaration by Researcher

I have provided the above participant with "Participant Information Sheet (PR/002)", given the participant the opportunity to ask further questions, and believe that the participant has understood the process.

Name of the Researcher (please print): **Asa Dorville**

Date (DD/MM/YYYY): **23/07/2026**

Participant Consent Form (PR/001):

Project Title: BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system

Declaration by Participant

Please tick to confirm:

☒ I have read the **Participant Information Sheet (PR/002)**, overleaf, and consent to take part in **BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system** by submitting both an image of a essay written in my handwriting and an audio recording of my voice reading that same essay.

If you are happy to be identified in materials (see PR/002) relating to you participation, please tick each of the options that you agree to:

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- ☒ I consent to being **referred to by my position / title**

If you **wish to be fully anonymised** in any such materials, please instead tick here only: ☐

Name of Participant (please print): Della Orniville

Contact details of the Participant: Email ellaornivilla@gmail.com Phone 1 758 729 6442

Date (DD/MM/YYYY): 5th/08/26

Declaration by Researcher

I have provided the above participant with "Participant Information Sheet (PR/002)", given the participant the opportunity to ask further questions, and believe that the participant has understood the process.

Name of the Researcher (please print): **Asa Dorville**

Date (DD/MM/YYYY): **23/07/2026**

Participant Consent Form (PR/001):

Project Title: BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system

Declaration by Participant

Please tick to confirm:

☒ I have read the **Participant Information Sheet (PR/002)**, overleaf, and consent to take part in **BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system** by submitting both an image of a essay written in my handwriting and an audio recording of my voice reading that same essay.

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- ☒ I consent to being **referred to by my position / title**

If you **wish to be fully anonymised** in any such materials, please instead tick here only: ☐

Name of Participant (please print): Gordon John

Contact details of the Participant: Email gordon.john@ccss.edu.lc Phone (758)72219650

Date (DD/MM/YYYY): 07/08/26

Declaration by Researcher

I have provided the above participant with "Participant Information Sheet (PR/002)", given the participant the opportunity to ask further questions, and believe that the participant has understood the process.

Name of the Researcher (please print): **Asa Dorville**

Date (DD/MM/YYYY): **23/07/2026**

Participant Consent Form (PR/001):

Project Title: BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system

Declaration by Participant

Please tick to confirm:

☒ I have read the **Participant Information Sheet (PR/002)**, overleaf, and consent to take part in **BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system** by submitting both an image of a essay written in my handwriting and an audio recording of my voice reading that same essay.

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- ☒ I consent to being **referred to by my position / title**

If you **wish to be fully anonymised** in any such materials, please instead tick here only: ☐

Name of Participant (please print): Hannah Mathurin

Contact details of the Participant: Email Hannahmathurin1@gmail.co Phone +1 (758) 284-0795

Date (DD/MM/YYYY): 01/08/2026

Declaration by Researcher

I have provided the above participant with "Participant Information Sheet (PR/002)", given the participant the opportunity to ask further questions, and believe that the participant has understood the process.

Name of the Researcher (please print): **Asa Dorville**

Date (DD/MM/YYYY): **23/07/2026**

Participant Consent Form (PR/001):

Project Title: BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system

Declaration by Participant

Please tick to confirm:

☒ I have read the **Participant Information Sheet (PR/002)**, overleaf, and consent to take part in **BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system** by submitting both an image of a essay written in my handwriting and an audio recording of my voice reading that same essay.

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☒ I consent to being **referred to by my position / title**

If you **wish to be fully anonymised** in any such materials, please instead tick here only: ☐

Name of Participant (please print): kaelum Prospere

Contact details of the Participant: Email kaelumpro10@gmail.com Phone 2859344

Date (DD/MM/YYYY): 31/07/2026

Declaration by Researcher

I have provided the above participant with "Participant Information Sheet (PR/002)", given the participant the opportunity to ask further questions, and believe that the participant has understood the process.

Name of the Researcher (please print): **Asa Dörville**

Date (DD/MM/YYYY): **23/07/2026**

Participant Consent Form (PR/001):

Project Title: BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system

Declaration by Participant

Please tick to confirm:

☒ I have read the **Participant Information Sheet (PR/002)**, overleaf, and consent to take part in **BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system** by submitting both an image of a essay written in my handwriting and an audio recording of my voice reading that same essay.

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- ☒ I consent to being **referred to by my position / title**

If you **wish to be fully anonymised** in any such materials, please instead tick here only: ☐

Name of Participant (please print): Kaylee Rickaille

Contact details of the Participant: Email Khailebailey@gmail.com Phone 17587299531

Date (DD/MM/YYYY): 30/07/2026

Declaration by Researcher

I have provided the above participant with "Participant Information Sheet (PR/002)", given the participant the opportunity to ask further questions, and believe that the participant has understood the process.

Name of the Researcher (please print): **Asa Dorville**

Date (DD/MM/YYYY): **23/07/2026**

Participant Consent Form (PR/001):

Project Title: BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system

Declaration by Participant

Please tick to confirm:

☒ I have read the Participant Information Sheet (PR/002), overleaf, and consent to take part in BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system by submitting both an image of a essay written in my handwriting and an audio recording of my voice reading that same essay.

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- ☐ I consent to being referred to by my position / title

If you wish to be fully anonymised in any such materials, please instead tick here only: ☒

Name of Participant (please print): Keishon Rushed Blummer

Contact details of the Participant: Email matthewedwards14@gmail.com Phone 722-5365

Date (DD/MM/YYYY): 12th/08/2026

Declaration by Researcher

I have provided the above participant with "Participant Information Sheet (PR/002)", given the participant the opportunity to ask further questions, and believe that the participant has understood the process.

Name of the Researcher (please print): **Asa Dorville**

Date (DD/MM/YYYY): **23/07/2026**

Participant Consent Form (PR/001):

Project Title: BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system

Declaration by Participant

Please tick to confirm:

☒ I have read the **Participant Information Sheet** (PR/002), overleaf, and consent to take part in **BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system** by submitting both an image of a essay written in my handwriting and an audio recording of my voice reading that same essay.

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- ☒ I consent to being **referred to by my position / title**

If you **wish to be fully anonymised** in any such materials, please instead tick here only: ☐

Name of Participant (please print): Krista Placide

Contact details of the Participant: Email kristayepacide@gmail.com Phone (758)286-7822

Date (DD/MM/YYYY): 12/04/2010

Declaration by Researcher

I have provided the above participant with "Participant Information Sheet (PR/002)", given the participant the opportunity to ask further questions, and believe that the participant has understood the process.

Name of the Researcher (please print): **Asa Dorville**

Date (DD/MM/YYYY): **23/07/2026**

Participant Consent Form (PR/001):

Project Title: BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system

Declaration by Participant

Please tick to confirm:

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If you **wish to be fully anonymised** in any such materials, please instead tick here only: ☒

Name of Participant (please print): Linsey Auguste

Contact details of the Participant: Email linsey.auguste@ccss.edu.lc Phone 17587292017

Date (DD/MM/YYYY): 31/07/2026

Declaration by Researcher

I have provided the above participant with "Participant Information Sheet (PR/002)", given the participant the opportunity to ask further questions, and believe that the participant has understood the process.

Name of the Researcher (please print): **Asa Dorville**

Date (DD/MM/YYYY): **23/07/2026**

Participant Consent Form (PR/001):

Project Title: BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system

Declaration by Participant

Please tick to confirm:

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If you **wish to be fully anonymised** in any such materials, please instead tick here only: ☒

Name of Participant (please print): Lydia Bailey

Contact details of the Participant: Email Lydicboiley300@gmail.com Phone 17585201980

Date (DD/MM/YYYY): 07/08/2026

Declaration by Researcher

I have provided the above participant with "Participant Information Sheet (PR/002)", given the participant the opportunity to ask further questions, and believe that the participant has understood the process.

Name of the Researcher (please print): **Asa Dorville**

Date (DD/MM/YYYY): **23/07/2026**

Participant Consent Form (PR/001):

Project Title: BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system

Declaration by Participant

Please tick to confirm:

☒ I have read the **Participant Information Sheet (PR/002)**, overleaf, and consent to take part in **BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system** by submitting both an image of a essay written in my handwriting and an audio recording of my voice reading that same essay.

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- ☐ I consent to being **referred to by my position / title**

If you **wish to be fully anonymised** in any such materials, please instead tick here only: ☐

Name of Participant (please print): Michia Leonce

Contact details of the Participant: Email michia.leonce@ccss.edu.lc Phone 719-0822

Date (DD/MM/YYYY): 31/07/2026

Declaration by Researcher

I have provided the above participant with "Participant Information Sheet (PR/002)", given the participant the opportunity to ask further questions, and believe that the participant has understood the process.

Name of the Researcher (please print): **Asa Dorville**

Date (DD/MM/YYYY): **23/07/2026**

Participant Consent Form (PR/001):

Project Title: BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system

Declaration by Participant

Please tick to confirm:

☒ I have read the **Participant Information Sheet (PR/002)**, overleaf, and consent to take part in **BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system** by submitting both an image of a essay written in my handwriting and an audio recording of my voice reading that same essay.

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- ☒ I consent to being **referred to by my position / title**

If you **wish to be fully anonymised** in any such materials, please instead tick here only: ☐

Name of Participant (please print): NASRI URIAH AUGUSTIN

Contact details of the Participant: Email nasriaugustin@gmail.com Phone +17587189955

Date (DD/MM/YYYY): 31/07/2026

Declaration by Researcher

I have provided the above participant with "Participant Information Sheet (PR/002)", given the participant the opportunity to ask further questions, and believe that the participant has understood the process.

Name of the Researcher (please print): **Asa Dorville**

Date (DD/MM/YYYY): **23/07/2026**

Participant Consent Form (PR/001):

Project Title: BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system

Declaration by Participant

Please tick to confirm:

☒ I have read the **Participant Information Sheet** (PR/002), overleaf, and consent to take part in **BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system** by submitting both an image of a essay written in my handwriting and an audio recording of my voice reading that same essay.

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☒ I consent to being **referred to by my position / title**

If you **wish to be fully anonymised** in any such materials, please instead tick here only: ☐

Name of Participant (please print): Neilaje Benjamin

Contact details of the Participant: Email whateverterr0r@hotmail.com Phone +1 (758) 727-0003

Date (DD/MM/YYYY): 12/08/2026

Declaration by Researcher

I have provided the above participant with "Participant Information Sheet (PR/002)", given the participant the opportunity to ask further questions, and believe that the participant has understood the process.

Name of the Researcher (please print): **Asa Dorville**

Date (DD/MM/YYYY): **23/07/2026**

Participant Consent Form (PR/001):

Project Title: BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system

Declaration by Participant

Please tick to confirm:

- ☒ I have read the **Participant Information Sheet** (PR/002), overleaf, and consent to take part in **BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system** by submitting both an image of a essay written in my handwriting and an audio recording of my voice reading that same essay.

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- ☒ I consent to being **referred to by my position / title**

If you **wish to be fully anonymised** in any such materials, please instead tick here only: ☐

Trisha Isaac on behalf of

Name of Participant (please print): Nubia Isaac

Contact details of the Participant: Email Simplynubia.8@gmail.com Phone 17587123886

Date (DD/MM/YYYY): _____

08/11/2010

Declaration by Researcher

I have provided the above participant with "Participant Information Sheet (PR/002)", given the participant the opportunity to ask further questions, and believe that the participant has understood the process.

Name of the Researcher (please print): **Asa Dorville**

Date (DD/MM/YYYY): **23/07/2026**

Participant Consent Form (PR/001):

Project Title: BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system

Declaration by Participant

Please tick to confirm:

☒ I have read the **Participant Information Sheet (PR/002)**, overleaf, and consent to take part in **BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system** by submitting both an image of a essay written in my handwriting and an audio recording of my voice reading that same essay.

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- ☒ I consent to being **referred to by my position / title**

If you **wish to be fully anonymised** in any such materials, please instead tick here only: ☐

Name of Participant (please print): Shamar Paul

Contact details of the Participant: Email _____ Phone 284 3534

Date (DD/MM/YYYY): 16/8/26

A. Dorville

Declaration by Researcher

I have provided the above participant with "Participant Information Sheet (PR/002)", given the participant the opportunity to ask further questions, and believe that the participant has understood the process.

Name of the Researcher (please print): **Asa Dorville**

Date (DD/MM/YYYY): **23/07/2026**

Participant Consent Form (PR/001):

Project Title: BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system

Declaration by Participant

Please tick to confirm:

☒ I have read the **Participant Information Sheet (PR/002)**, overleaf, and consent to take part in **BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system** by submitting both an image of a essay written in my handwriting and an audio recording of my voice reading that same essay.

If you are happy to be identified in materials (see PR/002) relating to you participation, please tick each of the options that you agree to:

- ☒ I consent to being **referred to by name**
- ☒ I consent to being **referred to by my place of work**
- ☒ I consent to being **referred to by my position / title**

If you **wish to be fully anonymised** in any such materials, please instead tick here only: ☐

Name of Participant (please print): Tamera Paul

Contact details of the Participant: Email _____ Phone 284 3534

Date (DD/MM/YYYY): 16/8/26 J. A. Aho

Declaration by Researcher

I have provided the above participant with "Participant Information Sheet (PR/002)", given the participant the opportunity to ask further questions, and believe that the participant has understood the process.

Name of the Researcher (please print): **Asa Dorville**

Date (DD/MM/YYYY): **23/07/2026**

Participant Consent Form (PR/001):

Project Title: BrainPower- A hybrid, multi-modal, accessible, Automated Essay Scoring (AES) system

Declaration by Participant

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If you **wish to be fully anonymised** in any such materials, please instead tick here only: ☐

Name of Participant (please print): Tia Wallace

Contact details of the Participant: Email tiawallace87@gmail.com Phone 1 (758) 520-6364

Date (DD/MM/YYYY): 06/08/2026

Declaration by Researcher

I have provided the above participant with "Participant Information Sheet (PR/002)", given the participant the opportunity to ask further questions, and believe that the participant has understood the process.

Name of the Researcher (please print): **Asa Dorville**

Date (DD/MM/YYYY): **23/07/2026**

Testing BrainPower AES System

The purpose of this notebook is to:

- Test the ASR and OCR components to justify their inclusion in the AES pipeline to provide accessibility and inclusion
- Test the Transformer component to determine its suitability for providing assistance to human scorers

The Prompt

```
prompt = ""More and more people use computers, but not everyone agrees that this benefits society.
```

```
Those who support advances in technology believe that computers have a positive effect on people.
```

```
They teach hand-eye coordination, give people the ability to learn about faraway places and people,
```

```
and even allow people to talk online with other people. Others have different ideas. Some experts
```

```
are concerned that people are spending too much time on their computers and less time exercising,
```

```
enjoying nature, and interacting with family and friends.
```

```
Write a letter to your local newspaper in which you state your opinion on the effects computers have
```

```
on people. Persuade the readers to agree with you."
```

Rubric Guidelines

```
import pandas as pd
```

```
df = pd.read_csv("./data/rubric guidelines.csv", encoding="utf-8")
```

```
df[:]
```

Output:

	score	description	typical element 1	typical element 2	typical element 3	typical element 4
0	1	An undeveloped response that may take a positi...	Contains few or vague details.	Is awkward and fragmented.	May be difficult to read and understand.	May show no awareness of audience.
1	2	An under-developed response that may or may no...	Contains only general reasons with unelaborate...	Shows little or no evidence of organization.	May be awkward and confused or simplistic.	May show little awareness of audience.
2	3	A minimally-developed response that may take a...	Has reasons with minimal elaboration and more ...	Shows some organization.	May be awkward in parts with few transitions.	Shows some awareness of audience.
3	4	A somewhat-developed response that takes a pos...	Has adequately elaborated reasons with a mix o...	Shows satisfactory organization.	May be somewhat fluent with some transitional ...	Shows adequate awareness of audience.

	score	description	typical element 1	typical element 2	typical element 3	typical element 4
4	5	A developed response that takes a clear posi...	Has moderately well elaborated reasons with mo...	Exhibits generally strong organization.	May be moderately fluent with transitional lan...	May show a consistent awareness of audience.
5	6	A well-developed response that takes a clear a...	Has fully elaborated reasons with specific det...	Exhibits strong organization.	Is fluent and uses sophisticated transitional ...	May show a heightened awareness of audience.

Method for Feature Extraction and Accuracy Metrics

```
import nltk

nltk.download('punkt_tab')

from jiwer import wer, cer, process_words

import re

from pathlib import Path

import numpy as np

import copy


class EssayEditor: #4/aug/2026
    """This class normalizes a body of text and can display the body of text along with
    its word and sentence counts.
    """

    #REGEX - compile patterns and rules once for efficiency

    REPLACEMENTS = {
        '/': ' or ',
        'slash': 'or',
        '@caps1': 'at caps one',
        '@caps2': 'at caps one',
        '@date1': 'at date one',
        '@organization1': 'at organization one',
        '@organization2': 'at organization two',
        'globe(astronomy)': 'globe astronomy',
    }

    SYMBOL_PATTERN = "|".join(re.escape(key) for key in REPLACEMENTS.keys())

    COMPILED_SYMBOL_PATTERN = re.compile(SYMBOL_PATTERN)

    PUNCTUATION_RULE = re.compile(r"[^\s\w]")#find all characters that are not words or spaces

    SPACES_RULE = re.compile(r"\s+")#find all occurrences of multiple spaces
```

```

def __init__(self, text):
    self.text = text
    self.original_text=copy.copy(text)
    self.edit_log = []#stores record of successful exection
    self.is_normalised = False
    sentences = nltk.tokenize.sent_tokenize(self.text)
    sentence_count = len(sentences)
    #Taken prior to normalization which removes sentence punctuation
    self.sentence_count = sentence_count

def normalise(self):
    self.text = self.text.lower()
    self.edit_log.append("Essay converted to lower case successfully.")
    # Normalise spoken forms and symbol
    self.text = self.COMPILED_SYMBOL_PATTERN.sub(
        lambda match: self.REPLACEMENTS[match.group(0)],
        self.text
    )
    self.edit_log.append("Symbols and spoken forms normalized successfully.")
    #Clean text
    #Replace all non-words or non-spaces with an empty string
    self.text = self.PUNCTUATION_RULE.sub("", self.text)
    #Replace all spaces with a single space, then remove unwanted spaces
    self.text = self.SPACES_RULE.sub(" ", self.text).strip()
    self.edit_log.append("Cleaned- punctuation an unwanted spaces removed successfully.")
    self.is_normalised = True

def get_sentence_count(self):
    return self.sentence_count

def get_text(self) -> str:#use type hint
    return self.text

def get_word_count(self):
    words= nltk.tokenize.word_tokenize(self.text)

```

```

        words = [word for word in words if word.isalnum()]

        word_count= len(words)

        return word_count


def show_normalised_text(self):

    print(self.text)


def show_original_text(self):

    print(self.original_text)


def show_edit_log(self):

    if self.is_normalised:

        print("\nNormalised:")

        for i, log in enumerate(self.edit_log,1):

            print(f"{i}: {log}")

    else:

        print("\nEssay not normalised")


def get_files_metadata(path_to_folder):

    """This function returns a list of the paths, filenames and extensions for
    all files in a folder.

    Path must be imported from pathlib

    """

    files_metadata = []

    #Folder with audio files

    folder = Path(path_to_folder)

    for file in folder.iterdir():#Returns Path object

        if file.is_file():

            #Get file path

            path = str(file)

            #Get file extension

            extension = file.suffix

            #Get file name/student's name

            name = file.stem

            #Get file size in bytes convert to mb

```

```

        size = round(file.stat().st_size/(1024**2), 2)

        #Add to list
        files_metadata.append({
            "path":path,
            "extension": extension,
            "name":name,
            "size":size
        })

    return files_metadata

def calculate_stats(list_of_nums):
    """This function takes as agrument and used numpy to calculates,
    the mean, median, min, max, std_dev of a list. It returns a
    dict.
    """
    #Convert to numpy array
    vals= np.array(list_of_nums)
    stats = {
        #Calculate mean
        "mean": f"{vals.mean():.2%}",
        #Calculate median
        "median": f"{np.median(vals):.2%}",
        #Calculate min
        "min": f"{vals.min():.2%}",
        #Calculate max
        "max": f"{vals.max():.2%}",
        #Calculate std_dev
        "std_dev": f"{vals.std():.2%}"
    }
    return stats

# Calculate WER and CER -----adapted from ChapGPT
def calculate_error_rates(reference_essay, component_output):
    word_error_rate = wer(reference_essay, component_output)
    character_error_rate = cer(reference_essay, component_output)

```

```

return {
    "wer": word_error_rate,
    "cer": character_error_rate
}

```

Output:

```

[nltk_data] Downloading package punkt_tab to
[nltk_data]   C:\Users\cowse\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt_tab is already up-to-date!

```

Unit Testing

```

# UT001 - Importing the ASAP 1.0 dataset
df = pd.read_csv(
    "./data/asap/training_set_rel3.tsv",
    sep="\t",
    encoding="latin-1"
)
df.head()

```

Output (5 rows × 28 columns, first 9 columns shown):

essay_id	essay_set	essay	rater1_domain1	rater2_domain1	rater3_domain1	domain1_score	rater1_domain2	rater2_domain2
0	1	1	Dear local newspaper, I think effects computer...	4	4	NaN	8	NaN
1	2	1	Dear @CAPS1 @CAPS2, I believe that using compu...	5	4	NaN	9	NaN
2	3	1	Dear, @CAPS1 @CAPS2 @CAPS3 More and more peopl...	4	3	NaN	7	NaN
3	4	1	Dear Local Newspaper, @CAPS1 I have found that...	5	5	NaN	10	NaN
4	5	1	Dear @LOCATION1, I know having computers has a...	4	4	NaN	8	NaN

```

#Get structure of dataset
df.columns.tolist()

```

Output:

```

['essay_id',
 'essay_set',
 'essay',
 'rater1_domain1',
 'rater2_domain1',
 'rater3_domain1',
 'domain1_score',
 'rater1_domain2',

```

```

'rater2_domain2',
'domain2_score',
'rater1_trait1',
'rater1_trait2',
'rater1_trait3',
'rater1_trait4',
'rater1_trait5',
'rater1_trait6',
'rater2_trait1',
'rater2_trait2',
'rater2_trait3',
'rater2_trait4',
'rater2_trait5',
'rater2_trait6',
'rater3_trait1',
'rater3_trait2',
'rater3_trait3',
'rater3_trait4',
'rater3_trait5',
'rater3_trait6']
#Get Statistics
df["essay_set"].value_counts().sort_index()

```

Output:

```

essay_set
1    1783
2    1800
3    1726
4    1770
5    1805
6    1800
7    1569
8     723

```

Name: count, dtype: int64

Extracting and Displaying Reference Essay

```

# UT002 - Extracting the first essay for the first prompt, then displaying it

#Extract essay
prompt1 = df[df["essay_set"] == 1].copy()
first_essay = prompt1.iloc[0]

```

```
reference_essay = first_essay["essay"]

#Create EssayEditor object
reference_essay = EssayEditor(reference_essay)

# Display reference essay
reference_essay.show_original_text()
```

Output:

Dear local newspaper, I think effects computers have on people are great learning skills/affects because they give us time to chat with friends/new people, helps us learn about the globe(astronomy) and keeps us out of troble! Thing about! Dont you think so? How would you feel if your teenager is always on the phone with friends! Do you ever time to chat with your friends or buisness partner about things. Well now - there's a new way to chat the computer, theirs plenty of sites on the internet to do so: @ORGANIZATION1, @ORGANIZATION2, @CAPS1, facebook, myspace ect. Just think now while your setting up meeting with your boss on the computer, your teenager is having fun on the phone not rushing to get off cause you want to use it. How did you learn about other countrys/states outside of yours? Well I have by computer/internet, it's a new way to learn about what going on in our time! You might think your child spends a lot of time on the computer, but ask them so question about the economy, sea floor spreading or even about the @DATE1's you'll be surprise at how much he/she knows. Believe it or not the computer is much interesting then in class all day reading out of books. If your child is home on your computer or at a local library, it's better than being out with friends being fresh, or being perpressedured to doing something they know isnt right. You might not know where your child is, @CAPS2 forbidde in a hospital bed because of a drive-by. Rather than your child on the computer learning, chatting or just playing games, safe and sound in your home or community place. Now I hope you have reached a point to understand and agree with me, because computers can have great effects on you or child because it gives us time to chat with friends/new people, helps us learn about the globe and believe or not keeps us out of troble. Thank you for listening.

Reference Essay Feature Extraction

```
#Display Extracted Features - Word and Sentence Counts (from raw essay)

print(f"Number of words: {reference_essay.get_word_count()}")

print(f"Number of sentences : {reference_essay.get_sentence_count()}")
```

Output:

Number of words: 331

Number of sentences : 16

Calculating Character and Word Error Rates

```
#The essay below has been delebrately modified in a controlled manner to determine if
#the WER and CER calculations are accurate. The digit '5' has been affixed to the ends
#of each of the first 3 and last 3 words of the essay.

modified_essay = ""Dear5 local5 newspaper5, I think effects computers have on people are great
learning skills/affects

because they give us time to chat with friends/new people, helps us learn about the
globe(astronomy) and keeps us

out of troble! Thing about! Dont you think so? How would you feel if your teenager is always on
the phone with friends!
```

Do you ever time to chat with your friends or buisness partner about things. Well now - there's a new way to chat the

computer, theirs plenty of sites on the internet to do so: @ORGANIZATION1, @ORGANIZATION2, @CAPS1, facebook, myspace

ect. Just think now while your setting up meeting with your boss on the computer, your teenager is having fun on the

phone not rushing to get off cause you want to use it. How did you learn about other countrys/states outside of yours?

Well I have by computer/internet, it's a new way to learn about what going on in our time! You might think your child

spends a lot of time on the computer, but ask them so question about the economy, sea floor spreading or even about the

@DATE1's you'll be surprise at how much he/she knows. Believe it or not the computer is much interesting then in class

all day reading out of books. If your child is home on your computer or at a local library, it's better than being out

with friends being fresh, or being perpressured to doing something they know isnt right. You might not know where your

child is, @CAPS2 forbidde in a hospital bed because of a drive-by. Rather than your child on the computer learning,

chatting or just playing games, safe and sound in your home or community place. Now I hope you have reached a point

to understand and agree with me, because computers can have great effects on you or child because it gives us time to

chat with friends/new people, helps us learn about the globe and believe or not keeps us out of troble.

Thank you5 for5 listening5.""

```
#Calculate Error Rates
```

```
error_rates = calculate_error_rates(reference_essay.text, modified_essay)
```

```
print(f"Word Error Rate: {error_rates['wer']:.3f} \nCharacter Error  
Rate:{error_rates['cer']:.3f}")
```

Output:

Word Error Rate: 0.018

Character Error Rate:0.046

Normalized Reference Essay

```
normalised_essay = EssayEditor(reference_essay.text)
```

```
normalised_essay.normalise()
```

```
normalised_essay.show_normalised_text()
```

```
normalised_essay.show_edit_log()
```

Output:

dear local newspaper i think effects computers have on people are great learning skills or affects because they give us time to chat with friends or new people helps us learn about the globe astronomy and keeps us out of trouble thing about dont you think so how would you feel if your teenager is always on the phone with friends do you ever time to chat with your friends or buisness partner about things well now theres a new way to chat the computer theirs plenty of sites on the internet to do so at organization one at organization two at caps one facebook myspace ect just think now while your setting up meeting with your boss on the computer your teenager is having fun on the phone not rushing to get off cause you want to use it how did you learn about other countrys or states outside of yours well i have by computer or internet its a new way to learn about what going on in our time you might think your child spends a lot of time on the computer but ask them so question about the economy sea floor spreading or even about the at date ones youll be surprise at how much he or she knows believe it or not the computer is much interesting then in class all day reading out of books if your child is home on your computer or at a local library its better than being out with friends being fresh or being perpressedured to doing something they know isnt right you might not know where your child is at caps one forbidde in a hospital bed because of a driveby rather than your child on the computer learning chatting or just playing games safe and sound in your home or community place now i hope you have reached a point to understand and agree with me because computers can have great effects on you or child because it gives us time to chat with friends or new people helps us learn about the globe and believe or not keeps us out of trouble thank you for listening

Normalised:

- 1: Essay converted to lower case successfully.
- 2: Symbols and spoken forms normalized successfully.
- 3: Cleaned- punctuation an unwanted spaces removed successfully.

Component Testing

Whisper (ASR) - Selecting the best model and configuration through experimentation

```
import librosa
import os
from faster_whisper import WhisperModel
import time

experiments= [] #stores experiment data as dictinaries 7/08/2026
size_compute_type_pairs= [
    ('small', 'int8_float16'),
    ('small', 'float16'),
    ('base', 'int8_float16'),
    ('base', 'float16'),
    ('medium', 'int8_float16'),
    ('medium', 'float16'),
]

#Get file and metadata
file_metadata= get_files_metadata("./data/audio/test")
```

```

if file_metadata:
    print(f"{file_metadata[0]['path']} accessed successfully.")
else:
    print("Error accessing file.")

#Load audio file and set sample rate
audio, rate = librosa.load(
    file_metadata[0]["path"],
    sr=16000
)

#Get length of audio
duration = librosa.get_duration(y=audio, sr=rate)
if duration:
    print("Audio successfully loaded.")
else:
    print("Error loading audio.")

for j, (m_size, c_type) in enumerate(size_compute_type_pairs, 1):

    #Start recording model loading time
    start_time = time.time()

    #load a model
    model = WhisperModel(
        model_size_or_path=m_size,
        device="cuda",
        compute_type=c_type,
        cpu_threads=16,
    )

    #Calculate time taken to load model
    model_load_time = time.time() - start_time

    #Transcribe audio for beam sizes 1 to 5

```

```

for i in range(1, 6):
    start_time = time.time()
    segments, info = model.transcribe(
        audio,
        language="en",
        beam_size=i,
        vad_filter=True
    )
    #Assemble transcription
    whisper_output = " ".join([segment.text for segment in segments])

    #Calculate transcription time
    inference_time = time.time() - start_time

    #Convert transcription string to EssayEditor Object for normalisation
    whisper_output = EssayEditor(whisper_output)

    #Normalise Essay
    whisper_output.normalise()

    #Calculate Transcription Accuracy
    error_rates = calculate_error_rates(normalised_essay.text, whisper_output.text)

    #ASR - Output Evaluation
    wer_threshold = 0.099
    cer_threshold = 0.059
    #IF both WER and CER below threshold, evaluation passed else evaluation failed
    evaluation_status = "pass" if error_rates['wer'] <= wer_threshold and error_rates['cer']
<= cer_threshold else "fail"

    #Store the details of the experiment as a dict
    details = {
        "Model Size": m_size,
        "Compute Type": c_type,

```

```

        "CPU Threads": 16,
        "Beam Size": i,
        "VAD Filter": True,
        "File Size": file_metadata[0]["size"],
        "File Length": duration,
        "Model Load Time": round(model_load_time, 2),
        "Transcription Time": round(inference_time, 2),
        "Transcription": whisper_output.text,
        "WER": round(error_rates["wer"], 3),
        "CER": round(error_rates["cer"], 3),
        "Evaluation Status": evaluation_status
    }

    #Save experiment details to experiments list
    experiments.append(details)

    print(f"Experiment {j}/6 completed")

```

Output:

```

data\audio\test\Daniella.wav accessed successfully.
Audio successfully loaded.
Experiment 1/6 completed
Experiment 2/6 completed
Experiment 3/6 completed
Experiment 4/6 completed
Experiment 5/6 completed
Experiment 6/6 completed

```

Display results for configurations

```

#Display Output as a dataframe
df = pd.DataFrame(experiments)

df= df[["Model Size",
        "Compute Type",
        "Beam Size",
        "Model Load Time",
        "Transcription Time",
        "WER",
        "CER",
        "Evaluation Status"
    ]]

df.head()

```

Output:

Model Size	Compute Type	Beam Size	Model Load Time	Transcription Time	WER	CER	Evaluation Status
small	int8_float16	1	2.35	5.83	0.100	0.050	fail
small	int8_float16	2	2.35	6.31	0.111	0.057	fail
small	int8_float16	3	2.35	6.60	0.094	0.046	pass
small	int8_float16	4	2.35	6.89	0.092	0.044	pass
small	int8_float16	5	2.35	6.65	0.119	0.074	fail

#Experiments 24 and 28 achieved the same transcription accuracy. However they differed greatly in the time taken to load the model.

#In experiment 24, mixed-precision quantization - int8_float16 was used, along with a beam size of 5. This model took roughly 4

#seconds to transcribe the audio. The model in experiment 28, with beam size 4 and compute_type set to float16, however,

#took about 50% less time to load but was more than 3 times slower than the model in experiment 24. It is likely that the reduced

#precision of int8 which produced smaller numerical values, as compared to float16, reduced the computation load resulting in

#significantly less transcription time.

Display in ascending order of WER and CER accuracy

```
#Sort dataframe in ascending order
```

```
df_sorted = df.sort_values(by=["WER", "CER"], ascending = True)
```

```
#Show only columns needed for selecting model specification
```

```
df_sorted = df_sorted[["Model Size", "Compute Type", "Beam Size", "Model Load Time",  
"Transcription Time", "WER", "CER", "Evaluation Status"]]
```

```
df_sorted.head()
```

Output:

idx	Model Size	Compute Type	Beam Size	Model Load Time	Transcription Time	WER	CER	Evaluation Status
24	medium	int8_float16	5	4.10	14.10	0.089	0.038	pass
28	medium	float16	4	2.12	43.81	0.089	0.038	pass
22	medium	int8_float16	3	4.10	13.34	0.092	0.039	pass
29	medium	float16	5	2.12	41.47	0.092	0.039	pass
3	small	int8_float16	4	2.35	6.89	0.092	0.044	pass

Whisper (ASR) Batch Transcription

```
#The model configuration chosen from the results of the previous experiments is:
```

```
# model_size = medium, beam_size=5, compute_type=int8_float16
```

```
from faster_whisper import WhisperModel
```

```
import librosa
```

```

import time

#Get a list with all file paths, extensions, names in audio folder
files_metadata = get_files_metadata("./data/audio")

#Start recording model loading time
start_time = time.time()

#Load model
model = WhisperModel(
    model_size_or_path="medium",
    device="cuda",
    compute_type="int8_float16",
    cpu_threads=16,
)

#Calculate model load time
model_load_time = time.time() - start_time
print(f"Whisper loaded successfully in {round(model_load_time,2)} seconds.")

#10/8/2026 converted to object
batch_transcription= {
    "details": [],
    "wer_stats": {"mean":'', "median":'', "min":'', "max":'', "std_dev":'',},
    "cer_stats": {"mean":'', "median":'', "min":'', "max":'', "std_dev":'',},
    "eval_stats":{
        "wer passes": 0, "wer fails": 0, "wer pass percentage": 0, "wer fail percentage": 0,
        "cer passes": 0, "cer fails": 0, "cer pass percentage":0, "cer fail percentage":0,
        "total passes": 0, "total fails": 0, "total pass percentage": 0, "total fail
percentage": 0,
    }
}

#Start Transcription Timer
start_time = time.time()

#Transcribe batch of audio files
for file in files_metadata:

```

```

#Load audio file and set sample rate
audio, rate = librosa.load(file["path"], sr=16000)

#Get length of audio
duration = librosa.get_duration(y=audio, sr=rate)
segments, info = model.transcribe(
    audio,
    language='en',
    beam_size=5,
    vad_filter=True
)
#Assemble transcription
whisper_output = " ".join([segment.text for segment in segments])
#Convert to EssayEditor Object for normalisation
whisper_output = EssayEditor(whisper_output)

#Normalise Essay
whisper_output.normalise()

#Calculate Error Rates
error_rates = calculate_error_rates(normalised_essay.get_text(), whisper_output.get_text())

#ASR - Output Evaluation
wer_threshold = 0.090
cer_threshold = 0.050
#IF both WER and CER below threshold evaluation passed else evaluation failed
evaluation_status = "pass" if error_rates["wer"] <= wer_threshold and error_rates["cer"] <= cer_threshold else "fail"

#Update the pass and fail counts accordingly
if evaluation_status == "fail":#Either one or both wer/cer failed
    if error_rates["wer"] > wer_threshold and error_rates["cer"] <= cer_threshold: #wer failed, cer passed
        batch_transcription["eval_stats"]["wer fails"] += 1
        batch_transcription["eval_stats"]["cer passes"] += 1

```

```

        elif error_rates["wer"] <= wer_threshold and error_rates["cer"] > cer_threshold: #wer
passed, cer failed

            batch_transcription["eval_stats"]["wer passes"] += 1

            batch_transcription["eval_stats"]["cer fails"] += 1

        else: #They both failed

            batch_transcription["eval_stats"]["cer fails"] += 1

            batch_transcription["eval_stats"]["wer fails"] += 1

        #Update total fail count

        batch_transcription["eval_stats"]["total fails"] += 1

    else: #They both passed

        batch_transcription["eval_stats"]["cer passes"] += 1

        batch_transcription["eval_stats"]["wer passes"] += 1

        batch_transcription["eval_stats"]["total passes"] += 1


#Store the details of the transcription as a dict
details = {

    "Student_ID": file["name"],

    "File Path": file["path"],

    "File Size": round(file['size'], 2),

    "File Length":round(duration,2),

    "File Extension":file["extension"],

    #decided to store the whole object whisper_output instead of just the transcription,
whisper_output.text

    #to all for inspection and analysis of both the original and normalised transcription

    "Transcription": whisper_output, #14/8/2026 passing object instead

    "WER": error_rates["wer"],

    "CER": error_rates["cer"],

    "Evaluation Status": evaluation_status

}

#Save transcription details to transcription list

batch_transcription["details"].append(details)


#End of transcription, details saved to list

inference_time = time.time() - start_time

print(f"Whisper successfully transcribed {len(batch_transcription['details'])} files in
{round(inference_time/60,2)} minutes.")

```

```

#Calculate Statistics for WER, CER

wer_list, cer_list=[],[]#Stores wer and cer statistics

for detail_obj in batch_transcription["details"]:
    wer_list.append(detail_obj["WER"])

wer_stats = calculate_stats(wer_list)

batch_transcription["wer_stats"].update(wer_stats)

print("WER statistics calculation complete.")


for detail_obj in batch_transcription["details"]:
    cer_list.append(detail_obj["CER"])

cer_stats = calculate_stats(cer_list)

batch_transcription["cer_stats"].update(cer_stats)

print("CER statistics calculation complete.")


#Calculating batch pass/fail statistics

num_transcriptions = len(batch_transcription['details'])

batch_transcription["eval_stats"]['total pass percentage'] =
f"{batch_transcription['eval_stats']['total passes'] / num_transcriptions: .2%}"

batch_transcription["eval_stats"]['total fail percentage'] =
f"{batch_transcription['eval_stats']['total fails'] / num_transcriptions: .2%}"

batch_transcription["eval_stats"]['wer pass percentage'] =
f"{batch_transcription['eval_stats']['wer passes'] / num_transcriptions: .2%}"

batch_transcription["eval_stats"]['wer fail percentage'] =
f"{batch_transcription['eval_stats']['wer fails'] / num_transcriptions: .2%}"

batch_transcription["eval_stats"]['cer pass percentage'] =
f"{batch_transcription['eval_stats']['cer passes'] / num_transcriptions: .2%}"

batch_transcription["eval_stats"]['cer fail percentage'] =
f"{batch_transcription['eval_stats']['cer fails'] / num_transcriptions: .2%}"

print("Evaluation statistics completed successfully.")

```

Output:

```

Whisper loaded successfully in 3.7 seconds.

Whisper successfully transcribed 27 files in 5.57 minutes.

WER statistics calculation complete.

CER statistics calculation complete.

Evaluation statistics completed successfully.

```

Batch Transcriptions Technical Details

```
#Display transcription data
```

```
df = pd.DataFrame(batch_transcription["details"], index=range(1,
len(batch_transcription["details"])+1))

df= df[[
    "Student_ID", "File Path", "File Size", "File Length", "WER", "CER", "Evaluation Status"
]]

df
```

Output:

Student_ID	File Path	File Size	File Length	WER	CER	Evaluation Status
Aaliah	data\audio\Aaliah.wav	10.60	115.82	0.080556	0.032551	pass
Alliah	data\audio\Alliah.wav	11.98	130.87	0.077778	0.019210	pass
Amanda	data\audio\Amanda.wav	26.82	146.45	0.130556	0.057631	fail
AnonOne	data\audio\AnonOne.wav	13.48	147.23	0.097222	0.043757	fail
AnonThree	data\audio\AnonThree.wav	12.95	141.45	0.069444	0.024013	pass
AnonTwo	data\audio\AnonTwo.wav	11.73	139.42	0.105556	0.035752	fail
Bevan	data\audio\Bevan.wav	12.69	138.55	0.111111	0.053895	fail
Brice	data\audio\Brice.wav	16.48	180.01	0.144444	0.062433	fail
Daniella	data\audio\Daniella.wav	16.83	183.77	0.088889	0.037887	pass
Darnel	data\audio\Darnel.wav	19.29	210.69	0.180556	0.109392	fail
Della	data\audio\Della.wav	10.87	118.72	0.155556	0.090181	fail
Gordon	data\audio\Gordon.wav	16.17	176.59	0.219444	0.131270	fail
Hannah	data\audio\Hannah.wav	12.01	131.18	0.130556	0.053895	fail
Kaelum	data\audio\Kaelum.wav	15.89	173.59	0.111111	0.048559	fail
Katelyn	data\audio\Katelyn.wav	22.41	133.24	0.147222	0.072572	fail
Kaylee	data\audio\Kaylee.wav	11.60	126.71	0.152778	0.069904	fail
Krista	data\audio\Krista.wav	14.30	170.02	0.069444	0.027215	pass
Michia	data\audio\Michia.wav	12.33	134.63	0.094444	0.046958	fail
Nasri	data\audio\Nasri.wav	15.45	168.75	0.122222	0.061366	fail
Neilage	data\audio\Neilage.wav	11.59	137.76	0.066667	0.024546	pass
Nubia	data\audio\Nubia.wav	12.13	132.46	0.102778	0.038954	fail
Reshan	data\audio\Reshan.wav	14.77	161.29	0.100000	0.036286	fail
Sheila	data\audio\Sheila.wav	15.85	173.08	0.100000	0.046958	fail
Shemar	data\audio\Shemar.wav	17.55	191.65	0.111111	0.052828	fail
Skyla	data\audio\Skyla.wav	15.33	167.49	0.088889	0.045891	pass
Tamera	data\audio\Tamera.wav	10.52	125.02	0.180556	0.120064	fail
Tia	data\audio\Tia.wav	10.60	115.82	0.080556	0.032551	pass

Descriptive Statistics

```
#Statistical Accuracy Statistics

transcription_statistics = {#10/08/2026

    "WER": batch_transcription["wer_stats"],

    "CER": batch_transcription["cer_stats"]

}

df = pd.DataFrame.from_dict(transcription_statistics)

df
```

Output:

	WER	CER
mean	11.55%	5.47%
median	10.56%	4.70%
min	6.67%	1.92%
max	21.94%	13.13%
std_dev	3.76%	2.82%

```
# Pass and Fail Statistics

df = pd.DataFrame.from_dict(

    batch_transcription['eval_stats'],orient="index", columns=["Batch Stats"])

df
```

Output:

	Batch Stats
wer passes	8
wer fails	19
wer pass percentage	29.63%
wer fail percentage	70.37%
cer passes	15
cer fails	12
cer pass percentage	55.56%
cer fail percentage	44.44%
total passes	8
total fails	19
total pass percentage	29.63%
total fail percentage	70.37%

Threshold Performance

```
batch_transcription["details"].sort(key=lambda item: item["WER"])
```

```
df = pd.DataFrame(batch_transcription["details"], index=range(1,
len(batch_transcription["details"])+1))

df= df[["Student_ID", "WER", "CER", "Evaluation Status"]]

df
```

Output:

Student_ID	WER	CER	Evaluation Status
Neilage	0.066667	0.024546	pass
AnonThree	0.069444	0.024013	pass
Krista	0.069444	0.027215	pass
Alliah	0.077778	0.019210	pass
Aaliah	0.080556	0.032551	pass
Tia	0.080556	0.032551	pass
Daniella	0.088889	0.037887	pass
Skyla	0.088889	0.045891	pass
Michia	0.094444	0.046958	fail
AnonOne	0.097222	0.043757	fail
Reshan	0.100000	0.036286	fail
Sheila	0.100000	0.046958	fail
Nubia	0.102778	0.038954	fail
AnonTwo	0.105556	0.035752	fail
Bevan	0.111111	0.053895	fail
Kaelum	0.111111	0.048559	fail
Shemar	0.111111	0.052828	fail
Nasri	0.122222	0.061366	fail
Amanda	0.130556	0.057631	fail
Hannah	0.130556	0.053895	fail
Brice	0.144444	0.062433	fail
Katelyn	0.147222	0.072572	fail
Kaylee	0.152778	0.069904	fail
Della	0.155556	0.090181	fail
Darnel	0.180556	0.109392	fail
Tamera	0.180556	0.120064	fail
Gordon	0.219444	0.131270	fail

Representative Transcriptions

```
#Calculate indices for 5 representative samples from WER distribution

#Select samples in increasing WER from min to max
```

```

min_idx = 0
max_idx = len(batch_transcription['details']) -1
median_idx = int((min_idx + max_idx)/2)
lower_mid_idx = int((min_idx + median_idx)/2)
upper_mid_idx = int((median_idx + max_idx)/2)

sample_idx = [min_idx, lower_mid_idx, median_idx, upper_mid_idx, max_idx]

for i in sample_idx:
    print(f"Student ID: {batch_transcription['details'][i]['Student_ID']}")
    print(f"WER: {batch_transcription['details'][i]['WER']:.2%}")
    print(f"CER: {batch_transcription['details'][i]['CER']:.2%}")
    print(f"Evaluation Status: {batch_transcription['details'][i]['Evaluation Status']}")
    batch_transcription['details'][i]["Transcription"].show_normalised_text()
    print('\n')

```

Output:

Student ID: Neilage

WER: 6.67%

CER: 2.45%

Evaluation Status: pass

dear local newspaper i think apex computers have on people are great learning skills or effects because they give us time to chat with friends or new people helps us learn about the globe astronomy and keeps us out of trouble think about dont you think so how would you feel if your teenager is always on the phone with friends do you ever tend to chat with your friends or base this partner about things well now theres a new way to chat the computer theres plenty of sites on the internet to do so at organization one at organization two at caps one facebook myspace etc just think now while youre setting up meeting with your boss on the computer your teenager is having fun on the phone not rushing to get off because you want to use it how did you learn about other countries or dates outside of yours well i have by computer or internet its a new way to learn about what going on in our time you might think your child spends a lot of time on the computer but ask them so question about the economy seafloor spreading or even about the at date once youll be surprised at how much he or she knows believe it or not the computer is much interesting than in class all day reading out of books if your child is home on your computer or at a local library its better than being out with friends being fresh or being peer pressured to doing something they know isnt right you might not know where your child is at caps two forbid in a hospital bed because of a driveby rather than your child on the computer learning chatting or just playing games safe and sound in your home or community place now i hope you have reached the point to understand and agree with me because computers can have great effects on you or child because it gives us time to chat with friends or new people helps us learn about the globe and believe or not keeps us out of trouble thank you for listening

Student ID: Daniella

WER: 8.89%

CER: 3.79%

Evaluation Status: pass

dear local newspaper i think effects computers have on people are great learning skills affects because they give us time to chat with friends new people helps us learn about the globe astronomy and keeps us out of trouble think about dont you think so how would you feel if your teenager is always on the phone with friends do you ever have time to chat with your friends or business partner about

things well now theres a new way to chat the computer theres plenty of sites on the internet to do so organization one at organization two at caps one facebook myspace etc just think now while youre setting up meeting with your boss on the computer your teenager is having fun on the phone not rushing to get off because you want to use it how did you learn about other countries states outside of yours well i have my computer internet its a new way to learn about what is going on in our time you might think your child spends a lot of time on the computer but ask them so questions about the economy see for spreading or even about the dates youll be surprised at how much she knows believe it or not the computer is much interesting than in class all day reading out of books if your child is home on your computer or at a local library its better than being out with friends being fresh or being pressured to do something they know isnt right you might not know where your child is at caps two forbid in a hospital bed because of a driveby rather than your child on the computer learning chatting or just playing games safe and sound in your home or community place now i hope you have reached a point to understand and agree with me because computers can have great effects on you or child because it gives us time to chat with friends new people helps us learn about the globe and believe or not keeps us out of trouble thank you for listening

Student ID: AnonTwo

WER: 10.56%

CER: 3.58%

Evaluation Status: fail

dear local newspaper i think effects computers have on people are great learning skills or effects because they give us time to chat with friends or new people help us learn about the globe astronomy and keep us out of trouble think about it dont you think so how would you feel if your teenager is always on the phone with friends do you ever have time to chat with your friends or business partners about things well now theres a new way to chat the computer theres plenty of sites on the internet to do so at organization one at organization two at caps one facebook myspace etc just think now while youre setting up meetings with your boss on the computer your teenager is having fun on the phone not rushing to get off because you want to use it how did you learn about other countries or states outside of yours well i have my computer or internet its a new way to learn about what is going on in our time you might think your child spends a lot of time on the computer but ask them some questions about the economy see floors spreading or even about the update ones youll be surprised at how much she or he knows believe it or not the computer is much interesting than in class all day reading out of books if your child is home on your computer or at a local library its better than being out with friends being fresh or being prepressured to do something they know isnt right you might not know where your child is at caps two forbid in a hospital bed because of a driveby rather than your child on a computer learning chatting or just playing games safe and sound in your home or community place now i hope you have reached a point to understand and agree with me because computers can have great effects on you and your child because it gives us time to chat with friends or new people help us learn about the globe and believe it or not keep us out of trouble thank you for listening

Student ID: Hannah

WER: 13.06%

CER: 5.39%

Evaluation Status: fail

dear local newspaper i think effects computers have on people are great learning skills or effects because they give us time to chat with friends or new people helps us learn about the globe astronomy and keeps us out of trouble think about it dont you think so how would you feel if your teenager is always on the phone with friends do you ever have time to chat with your friends or business partners about things well now theres a new way to chat the computer theres plenty of sites on the internet to do so at organization one as organization to add caps one facebook my space etc just think about while youre setting up a meeting with your boss on the computer your teenager is having fun on the phone not rushing to get off because you want to use it how did you learn about other countries or states outside of yours well i have by the computer or internet its a new way to learn about whats going on in our time you might think your child spends a lot of time on the computer but ask them some questions about the economy see flow spreading or even about the date youll be surprised at how much he or he knows believe it or not the computer is much more interesting than in class or the reading out of books if your child is home on your computer or at a local library its because them being out with friends being fresh or being peer pressured to do something they know isnt right you might not know where your child is at caps to forbid in a hospital bed because of a drive by rather than your child on the computer learning chatting or just playing games safe and sound in your home or community place now i have i hope you have reached a point to understand and agree with me because computers can have great effects on you or your child because it gives us time to chat with friends or new people helps us learn about the globe and believe it or not keeps us out of trouble thank you for listening

Student ID: Gordon

WER: 21.94%

CER: 13.13%

Evaluation Status: fail

they are local newspapers i think if you are from these foreign people i agree with learning skills or effects because they give us time to chat with friends or new people helps us learn about the global astronomy and keeps us out of trouble think about it dont you think so how would you feel if katinja is always on the phone with friends do you ever have time to chat with your friends or business partner about things well now theres a new way to chat no computer they spend their time on the internet to do so organization to capsy facebook microsoft etc just speak now while youre setting up meeting with your boss on the computer your teenager is having fun on the phone not rushing to get off because you want to use it how did you learn about other countries or dates outside of yours well i have my computer or internet its a new way to learn about whats going on in our time you might think your child spends a lot of time on the computer but ask them so so question about the economy seafloor spreading or even about the dates youll be surprised at how much easier she knows believe it or not the computer is much more interesting than in class all day reading out of books if your child is home on your computer or at the local library its better than being persisted to do something they know isnt right you might not know where your child is caps to forbidden in hospital bed because of a driveby rather than your child on the computer learning chat you know just playing games safe and sound in your home or community place now i hope you have reached a point to understand and agree with me because this is going to have great effects on you or child because it gives us time to chat with friends or new people helps us learn about the group and believe it or not gives us out of trouble thank you for listening

paddleOCR-OCR Model

```
from paddleocr import PaddleOCR

#You have to experiment to determine the best configuration like you did for Whisper

ocr = PaddleOCR(

    text_detection_model_name="PP-OCRv6_medium_det",

    text_recognition_model_name="PP-OCRv6_medium_rec",

    engine="transformers",

    use_doc_orientation_classify=True,

    use_doc_unwarping=True,

    use_textline_orientation=True,


    text_det_limit_side_len=text_det_limit_side_len,

    text_det_limit_type=text_det_limit_type,

    text_det_thresh=text_det_thresh,

    text_det_box_thresh=text_det_box_thresh,

    text_det_unclip_ratio=text_det_unclip_ratio,

    text_rec_score_thresh=text_rec_score_thresh,

    return_word_box=return_word_box,


    text_det_unclip_ratio=1.5, # default ~1.5; larger box expands region, helps if letters at
word edges get cut

)

print("PaddleOCR loaded successfully.")
```

Output:

Creating model: ('PP-LCNet_x1_0_doc_ori', None, 'transformers')

Model files already exist. Using cached files.

Loading weights: 100%

Creating model: ('UVDoc', None, 'transformers')

Model files already exist. Using cached files.

Loading weights: 100%

Creating model: ('PP-LCNet_x1_0_textline_ori', None, 'transformers')

Model files already exist. Using cached files.

Loading weights: 100%

Creating model: ('PP-OCRV6_medium_det', None, 'transformers')

Model files already exist. Using cached files.

Loading weights: 100%

Creating model: ('PP-OCRV6_medium_rec', None, 'transformers')

Model files already exist. Using cached files.

Loading weights: 100%

Text Detection, Recognition and Extraction

```
# Extract text

#1. call get_files_metadata()
#2. store file paths in a file_paths list
#3. use loop to get predictions
#4. store predictions in predictions list/dict
#5. Create a EssayEditor object for each prediction
#6. Normalise each prediction
#7. Calculate and store metrics for each prediction
#8. Calculate and store metrics for batch


#Get paths for all essays in sample
path_to_folder = "./data/essays"

files_list = get_files_metadata(path_to_folder)
print("Number of essays in sample:", len(path_list))


# Will store predictions for the batch
```

```

batch_extraction= { #TODO: Make into a class
    "details": [],
    "wer_stats": {"mean":'', "median":'', "min":'', "max":'', "std_dev":'',},
    "cer_stats": {"mean":'', "median":'', "min":'', "max":'', "std_dev":'',},
    "eval_stats":{
        "wer passes": 0, "wer fails": 0, "wer pass percentage": 0, "wer fail percentage": 0,
        "cer passes": 0, "cer fails": 0, "cer pass percentage":0, "cer fail percentage":0,
        "total passes": 0, "total fails": 0, "total pass percentage": 0, "total fail
percentage": 0,
    }
}

#Loop through files detecting and recognizing - predicting
for file in files_list:
    prediction = ocr.predict(file['path'])
    for result in prediction:
        sentences = result['rec_texts']
        #Combine into a single string
        essay = " ".join(sentence for sentence in sentences)
        #Convert to EssayEditor object
        paddleOCR_extraction = EssayEditor(essay)
        #Normalise extracted text
        paddleOCR_extraction.normalise()
        #Calculate Error Rates
        error_rates = calculate_error_rates(normalised_essay.get_text(),
paddleOCR_extraction.get_text())
        #OCR - Output Evaluation
        wer_threshold = 0.090
        cer_threshold = 0.050
        #IF both WER and CER below threshold evaluation passed else evaluation failed
        evaluation_status = "pass" if error_rates["wer"] <= wer_threshold and error_rates["cer"]
<= cer_threshold else "fail"

        #Update the pass and fail counts accordingly
        if evaluation_status == "fail":#Either one or both wer/cer failed
            if error_rates["wer"] > wer_threshold and error_rates["cer"] <= cer_threshold: #wer
failed, cer passed

```

```

        batch_extraction["eval_stats"]["wer fails"] += 1
        batch_extraction["eval_stats"]["cer passes"] += 1
        elif error_rates["wer"] <= wer_threshold and error_rates["cer"] > cer_threshold:
#wer passed, cer failed
            batch_extraction["eval_stats"]["wer passes"] += 1
            batch_extraction["eval_stats"]["cer fails"] += 1
        else: #They both failed
            batch_extraction["eval_stats"]["cer fails"] += 1
            batch_extraction["eval_stats"]["wer fails"] += 1
            batch_extraction["eval_stats"]["total fails"] += 1
    else: #They both passed
        batch_extraction["eval_stats"]["cer passes"] += 1
        batch_extraction["eval_stats"]["wer passes"] += 1
        batch_extraction["eval_stats"]["total passes"] += 1

confidence_list = []
for (text, score) in zip(result["rec_texts"], result["rec_scores"]):
    confidence_list.append(f"{score:.3f} | {text}")

#Store the details of the transcription as a dict
details = {
    "Student_ID": file["name"],
    "File Path": file["path"],
    "File Size": round(file['size'], 2),
    "File Extension":file["extension"],
    "Extraction": paddleOCR_extraction,
    "WER": error_rates["wer"],
    "CER": error_rates["cer"],
    "Confidence Scores": confidence_list,
    "Evaluation Status": evaluation_status
}

# Store in batch extractions
batch_extraction['details'].append(details)

```

Output:

Number of essays in sample: 27

#Calculate Batch Statistics for WER, CER

```

wer_list, cer_list=[],[]
for detail_obj in batch_extraction["details"]:
    wer_list.append(detail_obj["WER"])
    cer_list.append(detail_obj["CER"])

wer_stats = calculate_stats(wer_list)
batch_extraction["wer_stats"].update(wer_stats)
print("WER statistics calculation complete.")

cer_stats = calculate_stats(cer_list)
batch_extraction["cer_stats"].update(cer_stats)
print("CER statistics calculation complete.")

#Calculating batch pass/fail statistics
num_extractions = len(batch_extraction['details'])

batch_extraction["eval_stats"]['total pass percentage'] =
f"{batch_extraction['eval_stats']['total passes'] / num_extractions: .2%}"

batch_extraction["eval_stats"]['total fail percentage'] =
f"{batch_extraction['eval_stats']['total fails'] / num_extractions: .2%}"

batch_extraction["eval_stats"]['wer pass percentage'] = f"{batch_extraction['eval_stats']['wer
passes'] / num_extractions: .2%}"

batch_extraction["eval_stats"]['wer fail percentage'] = f"{batch_extraction['eval_stats']['wer
fails'] / num_extractions: .2%}"

batch_extraction["eval_stats"]['cer pass percentage'] = f"{batch_extraction['eval_stats']['cer
passes'] / num_extractions: .2%}"

batch_extraction["eval_stats"]['cer fail percentage'] = f"{batch_extraction['eval_stats']['cer
fails'] / num_extractions: .2%}"

print("Evaluation statistics completed successfully.")

```

Output:

```

WER statistics calculation complete.
CER statistics calculation complete.
Evaluation statistics completed successfully.

```

Threshold Performance

```

#Display transcription data
batch_extraction["details"].sort(key=lambda item: item["WER"])

df = pd.DataFrame(batch_extraction["details"], index=range(1,
len(batch_extraction["details"])+1))

df= df[["Student_ID", "WER", "CER", "Evaluation Status"]]

```

df

Output:

Student_ID	WER	CER	Evaluation Status
Nubia	0.111111	0.040021	fail
Skyla	0.172222	0.060832	fail
Tamera	0.200000	0.095518	fail
Anon 1	0.225000	0.130736	fail
Kaylee	0.230556	0.067236	fail
Neilage	0.236111	0.079509	fail
Shiela	0.238889	0.124333	fail
Anon 2	0.250000	0.120598	fail
Hannah	0.252778	0.167022	fail
Daniella	0.272222	0.075240	fail
Tia	0.330556	0.129136	fail
Alliah	0.333333	0.143543	fail
Della essay	0.333333	0.192636	fail
Aaliah	0.361111	0.189434	fail
Gordon essay	0.363889	0.130736	fail
Kaelum	0.366667	0.164888	fail
Amanda	0.375000	0.197439	fail
Michia	0.380556	0.274813	fail
Bevan	0.405556	0.177695	fail
Anon 3	0.413889	0.250267	fail
Krista	0.422222	0.240128	fail
Nasri	0.433333	0.242263	fail
Reshan	0.447222	0.160619	fail
Darnel	0.463889	0.266809	fail
Katelyn	0.472222	0.403415	fail
Brice	0.483333	0.251334	fail
Shemar	0.894444	0.751334	fail

Descriptive Statistics

#Statistical Accuracy Statistics

```
transcription_statistics = {#10/08/2026
    "WER": batch_extraction["wer_stats"],
    "CER": batch_extraction["cer_stats"]
}
```

```
df = pd.DataFrame.from_dict(transcription_statistics)
df
```

Output:

	WER	CER
mean	47.37%	20.64%
median	46.67%	18.36%
min	24.72%	5.44%
max	90.28%	75.93%
std_dev	11.61%	13.40%

```
# Pass and Fail Statistics
df = pd.DataFrame.from_dict(
    batch_extraction['eval_stats'],orient="index", columns=["Batch Stats"])
df
```

Output:

	Batch Stats
wer passes	0
wer fails	27
wer pass percentage	0
wer fail percentage	0
cer passes	1
cer fails	26
cer pass percentage	0
cer fail percentage	0
total passes	0
total fails	27
total pass percentage	0
total fail percentage	0

Representative Extractions

```
#Calculate indices for 5 representative samples from WER distribution
#Select samples in increasing WER from min to max
#TODO: Make all of this into a class
min_idx = 0
max_idx = len(batch_extraction['details']) -1
median_idx = int((min_idx + max_idx)/2)
lower_mid_idx = int((min_idx + median_idx)/2)
```

```

upper_mid_indx = int((median_indx + max_indx)/2)

sample_indx = [min_indx, lower_mid_indx, median_indx, upper_mid_indx, max_indx]

for i in sample_indx:

    print(f"Student ID: {batch_extraction['details'][i]['Student_ID']}")

    print(f"WER: {batch_extraction['details'][i]['WER']:.2%}")

    print(f"CER: {batch_extraction['details'][i]['CER']:.2%}")

    print(f"Evaluation Status: {batch_extraction['details'][i]['Evaluation Status']}")

    batch_extraction['details'][i]["Extraction"].show_normalised_text() #change Extraction to
Output for universality

    print('\n')

```

Output:

Student ID: Nubia

WER: 11.11%

CER: 4.00%

Evaluation Status: fail

dear local newspaper i think effects computers have on people are great learning skills or affects because they give us time to chat with friends or new people helps us learn about the global castronomy and keeps us out of troble thing about dont you think so how would you feel if your teenager is always on the phone with friends do you ever time to chat with your friends or buis partner about things well now theres a new way fo chat the computer theirs plenty of sites on the internet to do so organization1 organization 2 caps1 facebook myspace ect just think now while your setting up meeting with your bass on the computer your teenager is having fun on the phone not rushing to get off cause you want to use it tow did you learn about other countries or states outside of yours well i have by computer or internet its a new way to learn about what going on in eur time you might think your child speaks alot of the time on the computer but ask them so question about the economy sea floor spreading or even about the datels youll be surprised at how much he or she knows believe it or not the computer is much interesting then in class all day reading out of books if your child is home on your computer or at a local library its better than being out with friends being fresh or being perpressedured to do something they know ist right you might not know where your child is caps2 forbidde in the hospital bed because of a driveby rather than your child on a eomputer leerning chatting or just playing games safe and sound in your home or comntinity place now i hepe you have reached a point to understand and agree with me because computers can have great effects on you or child becaus it gives us time to chat with friends or new people help us learn about the globe and believe or not keeps us out of troble thank youfor listenina

Student ID: Shiela

WER: 23.89%

CER: 12.43%

Evaluation Status: fail

have on people are great learning skills affets beca they give us time to chat with friends or new peop helps us learn about the globe lastronomy and ke us out of troble thing about dont you think how would you foel if your teenager is always on th phone with friends do you ever time to chat with your friends or buisness parther about things we howtheres a new way to chat the computer theirs plenty of sites on the internet to do so at organization oneorgantzati 2at caps one facebook myspace ett just think now while your setting up meeting with your boss on the computer your teenas is having fun on the phone not rushing to got off cause you want to use it how did you learn about other countrys states outside of yours well i have by computer linternet its anew we to learn about what going on in our time youighyouo the computer but ask them so questron about exonomy sea flor spreading ox even about the at date ones youll be surprise at how much she knows befreve it or not the computer is much interesting then in class all day reading o of books if your child is home on your compute ata loa

library its batter than being out with friends being fresh or being perpressed to oing something they know isht right you might not know where your child is caps forbidde in a rospital bed begause of a driveby rather than your child on the computer learning chatiting or just playing games safe and sound in your home or community place now i hope you have reached a point to understand and agree with me beaause computers can have great effects on you or child because it gives us time to chat with friends new people alw us learn about the alobe and believe ox not

Student ID: Aaliah

WER: 36.11%

CER: 18.94%

Evaluation Status: fail

great leacning skills laffects because they give us time to chat with frienl new pegple helps us learn about the globe lastronomy and keepsus aut of troblething aboutoont you think so haw woaild you feel if your teenager is always on thephone with friends to yae ever time to chat with your friends or business partner about thinosiwell now thereis a new way to chat the compule theirs plenty of sites on the inlenet to do soorcanizationi at organization twocapsifocebookmyspocet just think now while your sething up meeting with your boss on the computkr your teenager is having fun on the phone not nushing to get off eause you want to use it how dil you learn about other countrys istades outside of yours well i have by computer or internetits a new way to learn about what going on n auetime you might think your child spends a ot of time on the computer but ask them so question about the econamysea floor spreading or even about the dateis youll be surprise at now much helshe knows believe it or not the computer is much interesting then in class all day reading out of books if your child is home on your computer or at a local libraryits bether than being out with friends being freshor being perpressed to doing something they know isnt right kou might not know where your child is capsz forbidde ina hospital bed because of a driveby rather than your child on the computer earningchatting or just playing games safe and sound in your home or community place now i hope pou have reached a point to understand and gee ithm bease compurnave great efedtna

Student ID: Anon 3

WER: 41.39%

CER: 25.03%

Evaluation Status: fail

deaa am ne people n lut af tin n wi nn ons nu fel l e on eev bou things well now theres a naw way to chat the cwmputer theirs plenty of sites on the internet to do so qrgrinization organizarion2 caps 1 faccbook my space etc just think now while your setting up meeting with your boss on the computeryour teenager is having fun on the phone not mushing to get off cause you warst to use it how did you learn about other countrms istates outside of yours dell i have by computer or internet its a nway to learn about whot guing on in our time you might think your child sperids a let of time on the computer but ash them so question about the economy sca floor spreadine or even about the dateis youll be surprise at how much helshe knows believe it or not the computer is much interesting then in class all doy reading out of books of your child is h on your computer or at a local library it better than being out with friends being fresh or being perpressed to doing something they thnow isrt right you might not thnow wherg your child is capsa forbidde in a hoppital bed becauss of a d bykathier than your child or the computer learningchatting or ust ploying gomes safe and sound in your home or commurn re vow i hope you have reached a point to understand and ag with me because computers can hore great effects on you or child because it gives us time to chat withfriendsinew pe helps us learn about the globe and believe or not heeps o out of trabk thank wou tor listening

Student ID: Shemar

WER: 89.44%

CER: 75.13%

Evaluation Status: fail

d or hc tdhy cxthmz x 2oson i h 4 entoret to a mes wey to learn annt s going on in ons timne you uight think var dhid sad alate of tame an the iter oat ash them 2o queton aront the ceaony flor soiredg or awen aut hudaks youll be aurpried at how much he or she iawbeliave tarnot anuter o mch intreoting n in dlars all dey reading ont of broods if your cild sit have a eth eth thy o ri hyaght t whear ehioe cena oh r sound in your hore or communy flace seao eeasena bee i e th t ri peohe thae or not bups o out o f froute phank yom cor loolag

Confidence Scores

```
#Lowest WER and CER scores
```

```
print("Student:", batch_extraction["details"][0]["Student_ID"])  
batch_extraction["details"][0]["Confidence Scores"]
```

Output:

Student: Nubia

```
['0.989 | Dear local newspaper, I think effects computers have on',  
'0.999 | people are great learning skills/affects because they give us',  
'0.998 | time to chat with friends/new people, helps us learn about the',  
'0.998 | global Castronomy) and keeps us out of troble! Thing about!'],  
'0.985 | Dont you think so? How would you feel if your teenager is',  
'0.996 | always on the phone with friends! Do you ever time to chat',  
'0.985 | with your friends or buis partner about things. well now-',  
"0.997 | there's a new way fo chat the computer, theirs plenty of sites",  
'0.960 | on the internet to do So: @ ORGANIZATION1, @ ORGANIZATION 2,',  
'0.965 | @ caps1, facebook, myspace ect.',  
'0.993 | Just think now while your setting up meeting with your bass',  
'0.996 | on the computer, your teenager is having fun on the phone not rushing',  
'0.982 | to get off cause you want to use it. tow did you learn about other',  
'0.988 | countries/states outside of yours? Well I have by computer/internet',  
"0.997 | it's a new way to learn about what going on in eur time!",  
'1.000 | You might think your child speaks alot of the time on the computer',  
'0.997 | but ask them so question about the economy, sea floor spreading or',  
"0.989 | even about the @ DATE1's you'll be surprised at how much he/she",  
'0.988 | knows: Believe it or not the computer is much interesting then in class',  
'0.987 | all day reading out of books: if your child is home on your computer',  
'0.985 | or at a local library, its better than being out with friends being',  
'1.000 | fresh or being perpressedured to do something they know ist right',  
'0.989 | You might not know where your child is @ caps2 forbidde in the',  
'0.982 | hospital bed because of a driveby, Rather than your child on a',  
'0.986 | eomputer leerning, chatting or just playing games. safe and sound in',  
'0.990 | your home or comntinity place.',  
'0.991 | Now I hepe you have reached a point to understand and agree with',  
'0.973 | me, because computers, can have great effects on you or child becaus.',  
'0.997 | it gives us time to chat with friends/new people, help us learn about',  
'1.000 | the globe and believe or not keeps us out of troble. Thank youfor',  
"0.937 | listenina."]
```

```
#Higest WER and CER scores
```

```
print("\nStudent:", batch_extraction["details"][26]["Student_ID"])  
batch_extraction["details"][26]["Confidence Scores"]
```

Output:

Student: Shemar

```
['0.644 | D/',  
 '0.388 | HC',  
 '0.535 | tDHy ',  
 '0.536 | cxthmz',  
 '0.677 | X',  
 '0.642 | 2oson@@',  
 '0.337 | I ',  
 '0.534 | H?',  
 '0.248 | 4',  
 '0.830 | entoret, to a mes wey to learn annt s going on in ons timne',  
 '0.717 | You ught think var dhid sad alate of tame an the iter oat ash them 2o queton aront the  
ceaony',  
 "0.666 | flor soredg or awen aut Hu@dak's you'll be aurpried at how much he/she iaw.Beliave tarnot",  
 '0.805 | anuter o mch intreoting n in dlars all dey reading ont of broods If your cild sit have a',  
 '0.411 | [,eth ',  
 '0.533 | eth thy o rihYaght t whear ehioe Cena oh',  
 '0.386 | R',  
 '0.923 | sound in your hore or communt y flace',  
 '0.440 | seao eeasena',  
 '0.517 | bee i e th t ri peohe thae',  
 '0.828 | or not bups o out of froute. Phank yom Cor loolag']
```

Appendix C

Implementation/Installation Requirements

#Generated by ClaudeAi, from the Jupyter Notebook

Core data handling

pandas>=2.0

numpy>=1.24

Text normalisation / tokenisation

nltk>=3.8

WER / CER scoring

jiwer>=3.0

ASR component

faster-whisper>=1.0

librosa>=0.10

OCR component (engine="transformers" requires paddlepaddle + transformers/torch backend)

paddleocr>=2.8

paddlepaddle>=2.6 # swap for paddlepaddle-gpu (matching your CUDA version) if using device="cuda"

transformers>=4.40

torch>=2.2

--- Notes for reviewers ---

1. nltk also needs the "punkt_tab" corpus downloaded at first run:

```
# python -m nltk.downloader punkt_tab
```

2. librosa requires the system binary "ffmpeg" to be installed and on PATH

(e.g. ``apt install ffmpeg`` / ``brew install ffmpeg`` / `choco install ffmpeg` on Windows).

3. faster-whisper and paddlepaddle were run with `device="cuda"` in this notebook,

which requires a working NVIDIA GPU + CUDA/cuDNN setup. To run on CPU instead,

change `device="cuda"` to `device="cpu"` in the `WhisperModel/PaddleOCR` calls

(expect much slower transcription/OCR).

4. Standard-library modules used (no install needed): `re`, `os`, `time`, `copy`, `pathlib`.